

# Summary of public comments

Considerations for the Regulation of Generative AI-Enabled Medical Devices  
Docket FDA-2026-N-7874

124 tagged comments, 2874 tags, 1668 skips, 5 duplicates not tagged · 30 sections · generated 06 October 2026

Each section says what commenters wrote about one part of FDA's paper. The line under a section's heading counts the comments and tags behind it, and the positions those tags take.

A number in brackets, such as [3], points to that section's sources, listed under its text. A source names a comment and a paragraph: 0126 ¶p091 is paragraph p091 of comment FDA-2026-N-7874-0126, so every claim can be looked up in the commenter's own words.

## GENERAL: Overall views, background and definitions

83 comments, 347 tags: 62 support, 211 conditional-minor, 34 conditional-major, 4 oppose, 36 neutral

### GEN: Overall approach and general views

77 comments, 267 tags: 55 support, 174 conditional-minor, 19 conditional-major, 3 oppose, 16 neutral

Many commenters welcome the paper's overall direction, often calling the two-axis risk framework, the competency-based approach or the life-cycle view sound; of the 267 points, 55 support FDA's approach as it stands and 174 ask for small changes [1].

The 19 points asking for larger changes put forward ideas, each from one commenter, such as licensure-like authorization in which autonomy is earned and authorization kept through continuing evidence [2], an execution-time gate so that an earlier authorization is not presumed to survive a material change [3], validating an evidence chain of deterministic checks and a qualified person's signature rather than the model [4], modular components as regulatory units [5], a partition that keeps a fallible model from writing the factual record [6], risk graded by whether a function generates or relays a clinical claim, and for whom [7], and clinical association put ahead of benchmarking [8].

Several stress least-burdensome, risk-proportionate oversight [9], and two want compliance costs not to shut out small or early-stage firms [10]. Others warn that flexibility must not outrun safety [11]: least-burdensome goals should not displace reasonable assurance [12], infrastructure such as trustworthy benchmarks should come first [13], monitoring and voluntary disclosure should prove reliable before earning reduced premarket evidence [14], pilot flexibility should depend on robust premarket evidence [15], and safety cannot rest on after-the-fact monitoring alone [16].

Several want existing machinery used rather than parallel tracks, such as ISO 14971 files and design, supplier and change controls, drug and biologic mechanisms such as established conditions, and existing classification regulations [17]; some add that GenAI need not displace established risk-based principles [18].

Several want evaluation to continue after authorization, treating it as an initial reliance boundary or adding continuous monitoring and requalification [19]; one would pair competency evidence with an auditable assurance case for the whole authorized system [20]. Single commenters also propose keeping competency, clinical confirmation and operational assurance as layers none can replace [21], keeping source evidence distinguishable from generated inference [22], separating capability from authority to act [23], and asking whether added evidence gives independent information about failure [24].

One commenter says accountability belongs to a person, so drafting and retrieval can stay light while consequential decisions need a signing clinician [25]; another wants claimed human oversight assessed against specific properties rather than credited as one category [26]; a third wants penalties, corrective action and enforcement when obligations are missed [27].

Other points ask FDA to pursue international harmonization, including through IMDRF [28]; to say which proposals need new legal authority or which mechanism applies [29]; to involve groups such as clinicians, patients and technical staff [30]; to cover lifecycle tasks such as operating, maintaining and retiring devices [31]; to address settings with no reachable clinician [32]; to report run-to-run variation with each performance estimate [33]; and to allow concluding that an intended use is not yet supportable [34].

The three opposing points find generative AI too risky for health care now, would bar it until open questions are settled, or would let market forces rein it in [35]. Neutral points include hope that generative AI could widen access, with safety first [36], and a warning that unconstrained generative AI places patients at risk of grievous harm [37].

**Sources.** [1] 0003 ¶p009, ¶p088; 0005 ¶p348; 0006 ¶p018; 0008 ¶p006; 0012 ¶p001; 0013 ¶p007; 0017 ¶p016; 0018 ¶p001; 0024 ¶p004; 0027 ¶p080; 0030 ¶p001; 0035 ¶p007; 0046 ¶p013; 0047 ¶p001; 0048 ¶p001; 0049 ¶p007; 0051 ¶p009; 0052 ¶p150; 0054 ¶p020; 0058 ¶p002; 0067 ¶p035; 0076 ¶p066; 0087 ¶p008; 0093 ¶p004; 0094 ¶p022; 0095 ¶p002; 0098 ¶p046; 0099 ¶p026; 0100 ¶p005; 0102 ¶p001; 0106 ¶p019; 0112 ¶p006; 0113 ¶p001; 0114 ¶p004; 0115 ¶p004; 0116 ¶p011; 0120 ¶p007; 0124 ¶p020 [2] 0045 ¶p012, ¶p013, ¶p100 [3] 0105 ¶p006, ¶p010 [4] 0103 ¶p009, ¶p010 [5] 0124 ¶p006 [6] 0127 ¶p013, ¶p027 [7] 0126 ¶p019 [8] 0113 ¶p047 [9] 0035 ¶p006; 0051 ¶p001; 0094 ¶p073; 0100 ¶p005; 0116 ¶p057; 0126 ¶p299 [10] 0003 ¶p083, ¶p084; 0094 ¶p020, ¶p022 [11] 0107 ¶p005; 0013 ¶p186; 0008 ¶p016; 0113 ¶p004; 0102 ¶p011 [12] 0107 ¶p005 [13] 0013 ¶p186 [14] 0008 ¶p016 [15] 0113 ¶p004 [16] 0102 ¶p011 [17] 0013 ¶p176; 0093 ¶p006, ¶p009; 0067 ¶p011, ¶p035; 0126 ¶p172 [18] 0030 ¶p016; 0074 ¶p007; 0087 ¶p050 [19] 0029 ¶p021; 0039 ¶p006; 0114 ¶p004; 0115 ¶p004 [20] 0035 ¶p005, ¶p129 [21] 0055 ¶p005, ¶p006, ¶p007, ¶p010 [22] 0054 ¶p022, ¶p100 [23] 0049 ¶p008 [24] 0104 ¶p006 [25] 0046 ¶p014, ¶p021 [26] 0098 ¶p003, ¶p026 [27] 0047 ¶p010 [28] 0003 ¶p087; 0024 ¶p010; 0038 ¶p005; 0051 ¶p009 [29] 0013 ¶p182; 0116 ¶p023 [30] 0017 ¶p054; 0038 ¶p006; 0032 ¶p022 [31] 0032 ¶p023 [32] 0082 ¶p011 [33] 0095 ¶p003 [34] 0013 ¶p184 [35] 0007 ¶p001; 0011 ¶p004; 0031 ¶p015, ¶p017 [36] 0086 ¶p002 [37] 0034 ¶p002

## BKG: GenAI characteristics, risks and benefits

19 comments, 26 tags: 5 support, 2 conditional-minor, 1 oppose, 18 neutral

Most points here take no position: 18 of the 26 offer evidence, examples, hopes or worries rather than a view on a proposal [1].

Five points agree with the paper's account of what sets GenAI apart, such as open-ended inputs, variable outputs and reliance on third-party models [2]; two add that this makes static or point-in-time validation insufficient, at least for some devices [3], and one that published studies suggest the paper reads, if anything, more confidently than the evidence supports [4].

Evidence of characteristic failures includes a chatbot suggesting bromide in place of salt [5], a research agent changing its action on identical reruns in 8.7 percent of test cells [6], a commercial assistant's transcriber producing fluent text after its audio input stopped [7], which one commenter says nonverbal users of communication devices could not catch or dispute [8], and studies linking training for user satisfaction to sycophancy [9].

Hoped-for benefits include saving lives, easing burnout and widening access [10], safer surgery and expert support beyond major centers [11], and helping clinicians stay current with evidence [12].

One point asking for a change says that GenAI usually fails not through single wrong outputs but through omission, harmful agreement, drift and escalation that fails in practice [13]; the other, that surgery, where actions may be hard to reverse and errors may have to be caught within seconds, needs explicit attention [14]. The opposing point warns of bias from non-representative training data [15].

**Sources.** [1] 0004 ¶p002; 0005 ¶p249; 0012 ¶p002; 0017 ¶p002, ¶p014; 0023 ¶p002, ¶p004; 0025 ¶p010, ¶p187, ¶p452; 0054 ¶p030; 0061 ¶p031; 0062 ¶p055; 0064 ¶p039; 0095 ¶p005, ¶p022; 0121 ¶p046 [2] 0020 ¶p011; 0029 ¶p004; 0038 ¶p001, ¶p004; 0106 ¶p011 [3] 0020 ¶p011; 0029 ¶p004 [4] 0106 ¶p011 [5] 0023 ¶p002 [6] 0095 ¶p022 [7] 0064 ¶p039 [8] 0062 ¶p055 [9] 0061 ¶p031, ¶p032 [10] 0012 ¶p002 [11] 0017 ¶p014 [12] 0025 ¶p187, ¶p452 [13] 0021 ¶p005 [14] 0017 ¶p018 [15] 0011 ¶p001

## DEF: Definitions, scope and device status

21 comments, 54 tags: 2 support, 35 conditional-minor, 15 conditional-major, 2 neutral

Two points back FDA's focus on intended use and function [1], one urging that regulation not follow the "generative AI" label [2]. Related requests: one standard whoever deploys a model [3], existing device

types kept for generative functions [4], device status set by the final function and deployment rather than latent capability alone [5], and one risk vocabulary across FDA centers for combination products [6].

Several want the line around low-risk functions drawn more clearly [7]: informational, non-directive functions treated as non-devices or under enforcement discretion [8], with its fit with the wellness policy clarified [9]; section 520(o)(1)(E)'s independent-review criterion used to mark lower-risk clinician-facing functions, which one commenter says generating functions generally cannot meet [10]; and a consistent line between AI that conveys a clinician's plan and AI exercising independent judgment [11]. One commenter instead wants the four-part decision support test re-examined, calling it seemingly arbitrary for modern chatbots [12].

The 15 points asking for larger changes include one commenter's multi-part request for premarket enforcement discretion when a generative draft report reaches only the interpreting radiologist, under conditions such as sequestration, local validation and monitoring [13]; output reaching clinicians or patients before specialist review would stay under premarket oversight [14].

Others worry about AI just outside the device boundary, such as administrative or agentic tools that shape when or where care happens, AI therapists claiming to treat psychiatric conditions, and upstream risks [15]. Proposals include an influence-based principle in which risk follows influence [16], an interagency framework [17], and a developer-to-user view modeled on farm-to-fork food safety [18].

Several want boundaries defined more carefully: where a function begins and ends, when local configuration needs review, how device and non-device functions on one platform are separated, and whether scoping to the GenAI component omits system issues such as servicing, or an implant's manufacturing handoff [19].

One commenter wants consensus definitions with agencies such as NIST, NIH and ONC [20]; another asks whether insight-generating devices count as generative [21].

**Sources.** [1] 0005 ¶p052; 0087 ¶p009 [2] 0087 ¶p009 [3] 0080 ¶p012 [4] 0126 ¶p249 [5] 0120 ¶p039, ¶p041 [6] 0067 ¶p013 [7] 0080 ¶p005; 0094 ¶p023; 0046 ¶p016; 0072 ¶p006 [8] 0080 ¶p005; 0094 ¶p023 [9] 0080 ¶p010 [10] 0046 ¶p016; 0126 ¶p087 [11] 0072 ¶p006 [12] 0116 ¶p020 [13] 0120 ¶p012, ¶p024, ¶p058, ¶p071, ¶p076 [14] 0120 ¶p013, ¶p104 [15] 0025 ¶p020, ¶p021; 0023 ¶p006; 0038 ¶p010 [16] 0025 ¶p440, ¶p442 [17] 0025 ¶p407 [18] 0038 ¶p011 [19] 0035 ¶p016, ¶p037; 0018 ¶p003, ¶p007; 0032 ¶p044; 0053 ¶p005 [20] 0038 ¶p009 [21] 0069 ¶p001

## RISK: Assessing a device's risk

72 comments, 521 tags: 47 support, 347 conditional-minor, 112 conditional-major, 7 oppose, 8 neutral

### Q1: Two-axis risk framework and additional dimensions

66 comments, 246 tags: 22 support, 129 conditional-minor, 92 conditional-major, 1 oppose, 2 neutral

Of the 246 points, 22 support the framework as it stands, 129 ask for small changes and 92 for larger ones. Many commenters call the two axes a sound or useful starting point [1].

Many propose adding the candidates FDA itself named: reversibility [2], time pressure or time to harm [3], traceability to primary sources [4] and, less often, downstream safeguards [5]. Several add whether the user can detect a wrong output [6]. Several also say human oversight should lower assessed risk only when it is meaningful rather than nominal, some asking that this be shown under realistic conditions [7]; one notes that state laws give the same supervision different force [8], and another wants review enforced by workflow controls distinguished from advisory review [9]. Some want a qualified person who approves or signs outputs to move a function down the risk scale [10].

Commenters split on form. Several would keep two axes and treat added factors as modifiers, to keep the grid usable [11]; others would add axes or layers, from four or five axes to a fifteen-driver inventory [12], or a further dimension for factors such as third-party model dependency, change dynamics or data exposure [13].

Two say the axes capture severity but not the probability of harm, and want probability or detectability handled as in established risk analysis [14]; one also wants the grid's relation to existing ISO 14971 risk files stated [15]. Another wants initial "criticality" kept distinct from product risk management [16], and one wants FDA's existing terms, model influence and decision consequence, reused [17]. Others say "incorrect output" misses harms from correct but incomplete answers, omission or harmful agreement [18].

Single commenters propose factors such as duration of reliance and the care available without the device [19], relational features and pediatric use [20], dependence on connectivity and shared infrastructure [21], whether records show who authorized an agent's action [22], implant geometry as physical output [23], deployment context such as missing specialist coverage [24], and whether a function relays a sourced claim or generates its own [25]. Two would let architecture, such as a deterministic interlock or limits on what the model can write, fix a function's place on the activity axis [26], and one would make irreversible actions ineligible for reduced premarket evidence [27]. Three favor anchoring safety-critical outputs in curated guidelines or fixed published protocols rather than open-ended generation [28].

Three want human or physician oversight kept over consequential AI decisions, one for any AI use [29], while two want uses such as ambient scribing or administrative tasks placed in lower-risk categories [30]. Two points agree that measurement and signal-processing functions may be higher risk because users cannot check their basis [31]. The one opposing point says the framework does not capture GenAI risk [32].

**Sources.** [1] 0003 ¶p016; 0009 ¶p008; 0015 ¶p001; 0020 ¶p012; 0028 ¶p002; 0029 ¶p019; 0030 ¶p002; 0035 ¶p039; 0038 ¶p012; 0043 ¶p002; 0046 ¶p002; 0049 ¶p024; 0051 ¶p010; 0052 ¶p004; 0054 ¶p054; 0058 ¶p032; 0076 ¶p024; 0080 ¶p004; 0087 ¶p011; 0093 ¶p010; 0094 ¶p033; 0099 ¶p007; 0100 ¶p021; 0102 ¶p005; 0103 ¶p011; 0112 ¶p007; 0115 ¶p008; 0118 ¶p007; 0120 ¶p129; 0121 ¶p034; 0127 ¶p002 [2] 0003 ¶p019; 0009 ¶p009; 0017 ¶p021; 0020 ¶p062; 0024 ¶p005; 0028 ¶p003; 0030 ¶p003; 0035 ¶p040;

0039 ¶p002; 0045 ¶p020; 0046 ¶p015; 0048 ¶p002; 0049 ¶p027; 0053 ¶p007; 0054 ¶p055; 0099 ¶p007; 0100 ¶p023; 0102 ¶p010; 0113 ¶p010; 0115 ¶p010; 0118 ¶p007; 0121 ¶p034; 0131 ¶p027 [3] 0003 ¶p020; 0009 ¶p009; 0015 ¶p001; 0017 ¶p021; 0028 ¶p005; 0035 ¶p041; 0045 ¶p021; 0054 ¶p055; 0082 ¶p016; 0086 ¶p003; 0113 ¶p010; 0115 ¶p010; 0118 ¶p010; 0119 ¶p028; 0131 ¶p029 [4] 0009 ¶p011; 0015 ¶p003; 0028 ¶p006; 0035 ¶p043; 0049 ¶p027; 0054 ¶p055; 0087 ¶p014; 0088 ¶p035; 0093 ¶p010; 0094 ¶p033; 0100 ¶p022; 0102 ¶p010; 0111 ¶p022; 0113 ¶p010; 0115 ¶p011; 0118 ¶p010; 0126 ¶p066 [5] 0028 ¶p004; 0030 ¶p003; 0049 ¶p027; 0052 ¶p024; 0082 ¶p016; 0111 ¶p042; 0115 ¶p014; 0131 ¶p029 [6] 0013 ¶p026; 0020 ¶p022; 0030 ¶p003; 0039 ¶p002; 0043 ¶p002; 0048 ¶p002; 0051 ¶p010; 0056 ¶p005; 0115 ¶p012 [7] 0005 ¶p108; 0009 ¶p010; 0017 ¶p047; 0021 ¶p016; 0025 ¶p400; 0055 ¶p031; 0058 ¶p035; 0087 ¶p012; 0098 ¶p028; 0099 ¶p013; 0120 ¶p050; 0121 ¶p084 [8] 0119 ¶p026 [9] 0120 ¶p133 [10] 0046 ¶p015; 0121 ¶p034; 0129 ¶p007 [11] 0028 ¶p021; 0035 ¶p047; 0049 ¶p024; 0051 ¶p010; 0055 ¶p021; 0087 ¶p015; 0131 ¶p024 [12] 0005 ¶p011; 0015 ¶p003; 0038 ¶p014; 0045 ¶p016; 0126 ¶p060 [13] 0003 ¶p018; 0093 ¶p005; 0076 ¶p032 [14] 0013 ¶p024, ¶p026; 0055 ¶p013, ¶p014 [15] 0013 ¶p028 [16] 0052 ¶p007 [17] 0067 ¶p017 [18] 0021 ¶p009; 0025 ¶p026, ¶p027; 0055 ¶p020; 0111 ¶p008 [19] 0006 ¶p019 [20] 0021 ¶p022, ¶p023 [21] 0032 ¶p020, ¶p052 [22] 0042 ¶p005 [23] 0053 ¶p007 [24] 0094 ¶p034 [25] 0126 ¶p066, ¶p067 [26] 0103 ¶p012; 0127 ¶p028, ¶p029 [27] 0131 ¶p027 [28] 0025 ¶p388; 0082 ¶p017; 0121 ¶p061 [29] 0023 ¶p006; 0047 ¶p003; 0097 ¶p013 [30] 0012 ¶p008; 0047 ¶p002 [31] 0049 ¶p025; 0126 ¶p053 [32] 0019 ¶p003

## Q2: Directiveness spectrum for informational functions

32 comments, 61 tags: 11 support, 40 conditional-minor, 9 conditional-major, 1 oppose

Of the 61 points, 11 support FDA's approach as it stands, 40 ask for small changes and 9 for larger ones. Many agree with FDA that directiveness turns on an output's substance and context rather than trigger words, that "talk to your doctor" disclaimers do not make an output less directive, or both [1].

Several propose modifiers such as specificity, personalization, imperative force, urgency, confidence presentation and whether alternatives are offered or dropped [2]. Several would judge directiveness by effect rather than text: what a reasonable user is likely to do, observed user behavior, how often clinicians override outputs, or influence on whether and where care is sought [3]; two want tests for directiveness that builds over a conversation or drifts beyond its intended level [4].

For predictability, several ask for worked examples, an anchored rubric, a decision tree or a table mapping modifiers to presumptive risk [5], and one for a rebuttable presumption that outputs with specific numbers, such as doses, are action-directing [6].

The points asking for larger changes propose other approaches, each from one commenter: a declared directiveness ceiling with a measured escape rate [7], one division between general and patient-specific output [8], an axis defined by the function's effect on the user's choice set, with a rung for options it omits [9], grading by significance because risk scores direct action at high values [10], or mandatory "for information only" wording on directive outputs not standardized and approved by FDA [11].

Two say greater directiveness is sometimes the safer design, as in emergencies, one asking for a "protocol delivery" sub-category judged on fidelity to published protocols [12]. Another argues that private drafts seen only by a radiologist sit at the low end despite diagnostic language [13]. The one opposing point wants a clear message on any action-directing medical guidance and continued reliance on human mental health professionals [14].

**Sources.** [1] 0006 ¶p020; 0020 ¶p013; 0028 ¶p022; 0038 ¶p017; 0049 ¶p030; 0055 ¶p030; 0076 ¶p024; 0082 ¶p018; 0093 ¶p011; 0100 ¶p024; 0113 ¶p012; 0118 ¶p012; 0120 ¶p137; 0126 ¶p049, ¶p051; 0127 ¶p020 [2] 0005 ¶p303; 0006 ¶p020; 0045 ¶p025; 0048 ¶p003; 0049 ¶p030; 0052 ¶p033, ¶p034; 0093 ¶p011; 0120 ¶p136 [3] 0025 ¶p059; 0048 ¶p003; 0049 ¶p031; 0058 ¶p037; 0113 ¶p012 [4] 0005 ¶p251; 0118 ¶p012 [5] 0006 ¶p020; 0028 ¶p022; 0051 ¶p011; 0093 ¶p011; 0100 ¶p025 [6] 0003 ¶p022 [7] 0013 ¶p031 [8] 0055 ¶p022, ¶p029 [9] 0131 ¶p004, ¶p033 [10] 0126 ¶p350 [11] 0043 ¶p007 [12] 0082 ¶p018, ¶p022; 0100 ¶p027 [13] 0120 ¶p137, ¶p138 [14] 0019 ¶p005

## Q3: Patient-facing vs clinician-facing functions

30 comments, 51 tags: 6 support, 38 conditional-minor, 5 conditional-major, 2 oppose

Of the 51 points, 6 support FDA's approach as it stands, 38 ask for small changes, 5 for larger ones and 2 oppose it. Many say patient-facing status should not by itself raise a function's risk [1]. Instead, they would weigh what an output could lead the user to do, and the stakes [2], whether the user can verify it in time or review can be enforced in the setting [3], and which safeguards the device demonstrates [4]. Several stress the access patient-facing tools offer or warn against paternalism [5].

Safeguards proposed include statements of uncertainty [6], traceable sources [7], escalation to care [8], comprehension testing across health-literacy levels [9], and a route to a human [10]. Three doubt that disclaimers or "consult your physician" lines protect users [11], while two suggest notices that outputs are not a diagnosis or a substitute for a physician [12].

Larger changes would rework the user variable: a separate user dimension, since users affect whether errors are caught, not their severity [13]; a third tier for supervised non-clinical health workers such as community health workers [14]; the verification path rather than the user's category [15]; or affirmation of the user's competency, required at higher risk, before a function unlocks [16]. Two argue that a clinical license does not make a user a reliable check on errors [17].

Single commenters suggest counting adaptation to a user's demonstrated understanding as a possible risk control [18], a "worker extender" certification tying a tool to a licensed worker's scope [19], fixed-rule selection of emergency protocols where no clinician is at hand [20], and keeping a radiologist's unverified AI pre-draft from patients and other clinicians [21]. One says patient-facing risk deserves more weight, since patients may not report or connect relevant symptoms and may use a tool instead of seeking care [22]. The opposing points call patient empowerment "a terrible goal" [23] and want patients to keep relying on human mental health professionals [24].

**Sources.** [1] 0003 ¶p024; 0006 ¶p021; 0013 ¶p032; 0028 ¶p009; 0030 ¶p005; 0035 ¶p051; 0037 ¶p008; 0045 ¶p026; 0049 ¶p032; 0051 ¶p012; 0054 ¶p057; 0076 ¶p065; 0100 ¶p028; 0113 ¶p013; 0115 ¶p015; 0120 ¶p141 [2] 0003 ¶p024; 0045 ¶p027; 0049 ¶p032; 0054 ¶p057; 0113 ¶p013; 0120 ¶p140, ¶p141 [3] 0013 ¶p033, ¶p034; 0052 ¶p040, ¶p048; 0115 ¶p015 [4] 0006 ¶p021; 0028 ¶p022; 0051 ¶p012; 0076 ¶p065; 0100 ¶p029, ¶p034 [5] 0006 ¶p021; 0021 ¶p024; 0055 ¶p033; 0100 ¶p028; 0126 ¶p098 [6] 0013 ¶p034; 0028 ¶p022; 0038 ¶p025; 0045 ¶p027; 0049 ¶p033; 0051 ¶p012; 0055 ¶p036; 0093 ¶p012; 0113 ¶p013 [7] 0013 ¶p034; 0037 ¶p008; 0051 ¶p012; 0055 ¶p035; 0100 ¶p030 [8] 0006 ¶p021; 0028 ¶p022; 0045 ¶p027; 0049 ¶p033; 0051 ¶p012; 0093 ¶p012; 0100 ¶p031; 0113 ¶p013 [9] 0005 ¶p304; 0006 ¶p021; 0028 ¶p022; 0051 ¶p012; 0055 ¶p039; 0093 ¶p012; 0100 ¶p033 [10] 0076 ¶p065; 0100 ¶p032 [11] 0006 ¶p021; 0037 ¶p008; 0052 ¶p047 [12] 0038 ¶p025; 0118 ¶p016 [13] 0126 ¶p097 [14] 0009 ¶p013, ¶p014 [15] 0013 ¶p034 [16] 0037 ¶p006, ¶p008 [17] 0025 ¶p067, ¶p068; 0037 ¶p008 [18] 0030 ¶p006 [19] 0094 ¶p037 [20] 0082 ¶p024, ¶p025 [21] 0120 ¶p142 [22] 0118 ¶p014, ¶p015 [23] 0043 ¶p009 [24] 0019 ¶p008

#### Q4: Generalist vs specialist clinician users

19 comments, 28 tags: 2 support, 23 conditional-minor, 1 conditional-major, 2 oppose

Of the 28 points, 2 support FDA's approach as it stands, 23 ask for small changes, 1 for a larger one and 2 oppose it. Several agree the distinction matters when safe use depends on specialist knowledge [1], but several would judge risk by whether the intended user can evaluate the output, not by title or category alone [2]. One opposing point would rest risk on intended use and its required qualifications instead [3]. Two give examples, from radiation oncology and mammography, of errors less specialized users may miss [4].

Several want sponsors to state the intended user's expected training or competency [5], and two would test against the least specialized intended user or generalist settings [6]. Four would test whether the device recognizes its limits and prompts specialist involvement [7]. Several want technical or workflow controls such as role-based access or review before an order [8], two doubting labeling alone [9]; one finds generic "consult a specialist" instructions inadequate for specific outputs [10], while another wants



outputs to tell generalists they are designed for specialists [11].

Four see an access benefit in bringing specialist knowledge to generalists [12], one warning that access to specialist information is not specialist competency [13]. One would also weigh whether errors are detectable and reversible and specialist escalation realistic [14]; another would credit specialist status only alongside conditions such as source access, sole control of finalization and training [15]. The other opposing point says only keeping humans in place can fix these risks [16].

**Sources.** [1] 0003 ¶p025; 0055 ¶p044; 0056 ¶p006; 0118 ¶p018; 0126 ¶p091; 0129 ¶p008 [2] 0005 ¶p305; 0028 ¶p022; 0049 ¶p034; 0052 ¶p050; ¶p052; 0056 ¶p006 [3] 0113 ¶p014 [4] 0056 ¶p007; 0126 ¶p091, ¶p092 [5] 0003 ¶p025; 0049 ¶p035; 0051 ¶p013; 0055 ¶p045; 0113 ¶p014; 0118 ¶p018 [6] 0051 ¶p013; 0093 ¶p013 [7] 0005 ¶p305; 0025 ¶p325; 0051 ¶p013; 0093 ¶p013 [8] 0013 ¶p036; 0049 ¶p035; 0055 ¶p048; 0056 ¶p009; 0113 ¶p014; 0129 ¶p008 [9] 0013 ¶p036; 0049 ¶p035 [10] 0056 ¶p008 [11] 0043 ¶p011 [12] 0025 ¶p323; 0049 ¶p034; 0051 ¶p013; 0093 ¶p013 [13] 0025 ¶p327 [14] 0056 ¶p008 [15] 0120 ¶p145, ¶p147 [16] 0019 ¶p012

## Q5: Multi-turn conversational risk

34 comments, 73 tags: 4 support, 63 conditional-minor, 2 conditional-major, 4 neutral

Of the 73 points, 4 support FDA's approach as it stands, 63 ask for small changes, 2 for larger ones and 4 take no position. Most agree that risk should be assessed across whole conversations rather than single turns [1], one calling this, as far as it knows, ahead of most regulatory thinking internationally [2]. Several say single-turn testing alone cannot show the risk [3].

For intended use, several would declare an envelope of scope, states, permitted transitions or forbidden behaviors that the device must hold [4]. Several would classify a device by the furthest point a conversation can plausibly reach, or is designed to reach, rather than its opening [5], while one says an out-of-scope capability should not by itself redefine intended use [6]. The two larger changes would let a clinical claim at any turn set the whole exchange's position [7], or characterize intended use by what architectural gates permit, turning migration into logged gate refusals [8].

Proposed tests include adversarial trajectories in which simulated users minimize symptoms, resist escalation, press for advice or try to bypass limits [9]; long conversations, new sessions and stored memory [10]; and missing, contradictory or changing information [11]. Several want the device to recognize drift and then stop, revalidate or escalate to a human [12], and several favor enforceable or deterministic controls [13], two preferring them to prompt instructions [14]. One would keep oversight high whenever a device can migrate [15]. One reports having independently identified cumulative "reassurance drift" and wants a separate test for it [16]; another reports safeguards eroding in longer conversations and wants months of evaluation for devices used daily [17].

One commenter describes one recorded session in which an assistant on a drafting task turned directive after a user's correction, acting on frustration it inferred but could not perceive, while its citations and self-corrections improved and its false statements continued [18]. That commenter would vary expressed emotion while holding the clinical request constant [19], count unrequested directives by turn [20], compare early and late turns [21], and name the trigger in the intended use [22], warning that distressed users produce it [23]. One wants a prespecified observation window, counting late escalation as a failure even if recovered [24], and another notes devices re-asking questions already answered, a failure to use the state already collected [25].

**Sources.** [1] 0003 ¶p026; 0005 ¶p306; 0006 ¶p022; 0013 ¶p038; 0017 ¶p023; 0020 ¶p014; 0025 ¶p100; 0028 ¶p010; 0035 ¶p048; 0045 ¶p030; 0049 ¶p036; 0052 ¶p053; 0055 ¶p053; 0065 ¶p013; 0076 ¶p025; 0105 ¶p022; 0113 ¶p015; 0114 ¶p010; 0115 ¶p016; 0118 ¶p020; 0120 ¶p149; 0127 ¶p021 [2] 0076 ¶p025 [3] 0013 ¶p039; 0055 ¶p053; 0098 ¶p032; 0114 ¶p011 [4] 0003 ¶p027; 0006 ¶p022; 0013 ¶p038; 0028 ¶p010; 0035 ¶p049; 0051 ¶p014; 0093 ¶p014; 0113 ¶p015 [5] 0003 ¶p027; 0052 ¶p053; 0065 ¶p034; 0118



¶p020; 0131 ¶p041 [6] 0120 ¶p150 [7] 0126 ¶p107 [8] 0127 ¶p031 [9] 0006 ¶p022; 0013 ¶p038; 0025 ¶p110; 0035 ¶p050; 0045 ¶p031; 0055 ¶p055; 0111 ¶p012; 0115 ¶p016; 0118 ¶p020 [10] 0021 ¶p014; 0055 ¶p054, ¶p058, ¶p059; 0065 ¶p039; 0105 ¶p024; 0115 ¶p016 [11] 0025 ¶p109; 0045 ¶p031; 0055 ¶p057; 0105 ¶p024; 0114 ¶p011; 0115 ¶p016 [12] 0003 ¶p027; 0043 ¶p013; 0049 ¶p037; 0105 ¶p023; 0113 ¶p015; 0120 ¶p151 [13] 0052 ¶p054; 0055 ¶p061, ¶p070; 0065 ¶p042; 0111 ¶p013; 0120 ¶p151 [14] 0065 ¶p042; 0111 ¶p013 [15] 0037 ¶p009 [16] 0098 ¶p031, ¶p032 [17] 0021 ¶p013, ¶p014 [18] 0065 ¶p016, ¶p022, ¶p025 [19] 0065 ¶p035 [20] 0065 ¶p036 [21] 0065 ¶p038 [22] 0065 ¶p043 [23] 0065 ¶p032 [24] 0091 ¶p040, ¶p041 [25] 0088 ¶p029

## Q6: Care escalation: under- vs over-escalation

36 comments, 62 tags: 2 support, 54 conditional-minor, 3 conditional-major, 1 oppose, 2 neutral

Of the 62 points, 2 support FDA's approach as it stands, 54 ask for small changes, 3 for larger ones, 1 opposes it and 2 take no position. Most want missed and unnecessary escalations measured and reported separately rather than folded into one accuracy score [1]. Many want sponsors to prespecify and justify, for their clinical context, how they weigh the two errors or where they set the operating point, rather than meet one universal threshold [2]; one warns that otherwise a sponsor can choose the more favorable result afterwards [3], and one would have independent experts chosen by professional societies set the criteria [4].

Several agree with FDA that the two errors are not commensurable [5], while one suggests treating them as sensitivity and specificity for each condition [6]. Several would give missed escalations more weight, or less tolerance, than unnecessary ones, especially in serious or emergency presentations such as alarm features or anaphylaxis [7]; one would reject emergency department cost arguments for looser patient-facing thresholds [8], and two would let a severe miss act as a pass-or-fail gate of its own [9]. Several also stress over-escalation's harms, such as anxiety, testing, cost, crowding and lost trust [10], or unwanted police responses [11]; one says these harms are under-detected by postmarket monitoring [12], and another that over-escalation can be the more dangerous error [13].

Two would also judge the level, timing and destination of escalation [14], two whether the handoff actually happens [15], and one, for emergency protocols, when the call for help comes [16]. The larger changes add a third error, escalating badly [17], routing to an accountable care team as a distinct option [18], and the severe-miss gate [19]. Two say an intermediate step such as a telehealth nurse, or a device that offers rather than initiates contact, makes over-escalation more acceptable or less severe [20].

Single commenters would also require corpora with declared non-escalation cases [21], weigh probability, severity and consequence of delay [22], or see detailed weighting as often possible only in product-specific risk management [23]. The opposing point sees no acceptable trade-offs, as AI now carries too much risk [24].

**Sources.** [1] 0005 ¶p308; 0006 ¶p023; 0009 ¶p021; 0017 ¶p023; 0020 ¶p016; 0035 ¶p050, ¶p051; 0045 ¶p033; 0049 ¶p039; 0051 ¶p015; 0054 ¶p062; 0067 ¶p014; 0091 ¶p044; 0093 ¶p015; 0100 ¶p037; 0111 ¶p016; 0113 ¶p016; 0115 ¶p016; 0116 ¶p052; 0118 ¶p022; 0120 ¶p153 [2] 0005 ¶p297; 0006 ¶p023; 0009 ¶p021; 0013 ¶p041; 0026 ¶p003; 0028 ¶p011; 0045 ¶p033; 0049 ¶p039; 0051 ¶p015; 0067 ¶p015; 0076 ¶p061; 0084 ¶p003; 0091 ¶p047; 0093 ¶p015; 0100 ¶p036; 0111 ¶p016; 0120 ¶p155 [3] 0013 ¶p041 [4] 0116 ¶p052 [5] 0006 ¶p023; 0013 ¶p040; 0052 ¶p060; 0091 ¶p044; 0093 ¶p015; 0100 ¶p035; 0111 ¶p016 [6] 0126 ¶p108 [7] 0003 ¶p028; 0006 ¶p023; 0026 ¶p002; 0054 ¶p048; 0076 ¶p061; 0100 ¶p037; 0118 ¶p022; 0120 ¶p155 [8] 0076 ¶p061 [9] 0005 ¶p308; 0055 ¶p083 [10] 0003 ¶p028; 0006 ¶p023; 0025 ¶p117; 0084 ¶p002; 0111 ¶p015 [11] 0021 ¶p015; 0111 ¶p015 [12] 0013 ¶p043 [13] 0043 ¶p015 [14] 0052 ¶p061, ¶p067; 0055 ¶p080 [15] 0021 ¶p016; 0055 ¶p076, ¶p081 [16] 0082 ¶p027 [17] 0021 ¶p015 [18] 0009 ¶p019, ¶p020 [19] 0055 ¶p083 [20] 0026 ¶p003; 0111 ¶p017 [21] 0091 ¶p046 [22] 0025 ¶p114 [23] 0052 ¶p060 [24] 0019 ¶p016

## PREMARKET: Evaluating a device before it is sold

82 comments, 921 tags: 94 support, 693 conditional-minor, 101 conditional-major, 7 oppose, 26 neutral

### Q7: Competency-based approach overall

50 comments, 126 tags: 40 support, 59 conditional-minor, 26 conditional-major, 1 neutral

Of the 126 points, 40 support FDA's approach as it stands, 59 ask for small changes, 26 for larger ones and 1 takes no position. Most accept benchmarking followed by clinical confirmation as a useful frame, often with conditions [1], several because open-ended outputs cannot be tested exhaustively [2]. Many endorse evaluating the final configured device rather than the foundation model alone [3], one calling this the paper's most important sentence [4]. Several want results tied to an identified, fixed configuration, such as model version, prompts, retrieval, tools and guardrails [5], and several would widen the unit to the sociotechnical system, the human-AI team in its workflow, a sensor pathway or, for implants, the validated production system [6].

Several caution that the clinician analogy carries over credentialing's structure but not the accountability or measurement properties behind it: devices face no license revocation or peer review, can be tested at scale and fail in correlated ways [7]. Two would let it shape the evidence but not reduce its amount or loosen change control [8]. Larger changes on accountability would add a fast process to suspend authorization on prespecified triggers, like revoking a license [9], or credential the whole system with responsibility allocated among manufacturer, institution and clinicians [10].

Several warn that competency may become a surrogate: benchmark success may not carry over to clinical practice or show clinical benefit [11]. Four call competency testing necessary but not sufficient, a gate rather than full assurance, or a supplement to software validation and postmarket accountability [12]. Several would scope competency to defined tasks and conditions [13], three counting recognition of limits, abstention or escalation as part of it [14]. Three would turn the analogy's supervised practice into staged deployment, widening scope only after evidence gates [15].

One commenter argues that open-endedness and output variability are largely design choices, so sponsors should show they apply, bounding inputs and outputs should be the primary expectation, validated in its own right, and conditions that cannot be evidenced should be excluded and deferred to humans [16]; it argues that competency inference is weakest for rare, high-consequence cases, while deferral can be tested directly [17]. Other larger changes would establish clinical association in a research study before benchmarking [18], make human factors validation a separate pillar [19], credential a deterministic gate and its signing person rather than the model [20], or set competency standards for research-site staff acting on AI outputs [21].

Single commenters also ask to separate output errors from their clinical consequences [22], disclose whether unverified streamed text reaches users [23], state in the authorization when inference is remote [24], include real patient conversations in training data [25], and test under pressure, hierarchy and unfamiliar patients, as real examinations do [26]. One asks how patient data would be protected [27].

**Sources.** [1] 0003 ¶p030; 0006 ¶p029; 0012 ¶p003; 0013 ¶p046; 0017 ¶p025; 0025 ¶p145; 0028 ¶p023; 0035 ¶p053; 0043 ¶p016; 0045 ¶p035; 0049 ¶p041; 0052 ¶p069; 0054 ¶p066; 0055 ¶p084; 0058 ¶p002; 0087 ¶p022; 0088 ¶p021; 0091 ¶p035; 0093 ¶p016; 0094 ¶p041; 0100 ¶p040; 0102 ¶p005; 0106 ¶p013; 0113 ¶p019; 0114 ¶p003; 0115 ¶p018; 0118 ¶p024; 0120 ¶p157; 0123 ¶p014; 0124 ¶p002 [2] 0006 ¶p029; 0028 ¶p023; 0052 ¶p069; 0054 ¶p066; 0093 ¶p016; 0102 ¶p005; 0114 ¶p003; 0123 ¶p014 [3] 0017 ¶p025; 0027 ¶p019; 0049 ¶p041; 0053 ¶p008; 0054 ¶p066; 0055 ¶p085; 0076 ¶p026; 0087 ¶p017; 0088 ¶p024; 0093 ¶p016; 0100 ¶p040; 0114 ¶p007; 0120 ¶p038; 0121 ¶p083; 0127 ¶p019 [4] 0076 ¶p026 [5] 0005 ¶p264; 0024 ¶p008; 0035 ¶p053; 0076 ¶p012;

0115 ¶p018; 0123 ¶p019 [6] 0017 ¶p015; 0053 ¶p008; 0054 ¶p042; 0074 ¶p014, ¶p016; 0123 ¶p020 [7] 0013 ¶p047, ¶p048; 0020 ¶p032, ¶p036; 0052 ¶p072; 0076 ¶p046; 0123 ¶p017 [8] 0013 ¶p051; 0076 ¶p051 [9] 0076 ¶p048 [10] 0123 ¶p020 [11] 0020 ¶p026, ¶p030; 0087 ¶p021; 0094 ¶p019; 0107 ¶p017, ¶p018; 0126 ¶p124 [12] 0049 ¶p042; 0055 ¶p086; 0118 ¶p026; 0123 ¶p016 [13] 0024 ¶p006; 0056 ¶p004; 0114 ¶p006; 0123 ¶p019 [14] 0025 ¶p147; 0056 ¶p004; 0114 ¶p005 [15] 0045 ¶p036; 0052 ¶p076, ¶p079; 0120 ¶p055, ¶p057 [16] 0126 ¶p121, ¶p122, ¶p138, ¶p139, ¶p140 [17] 0126 ¶p127, ¶p128 [18] 0113 ¶p019 [19] 0018 ¶p017 [20] 0103 ¶p014 [21] 0112 ¶p014 [22] 0052 ¶p073 [23] 0098 ¶p041 [24] 0079 ¶p073 [25] 0118 ¶p024 [26] 0058 ¶p053 [27] 0019 ¶p018

## Q8: Using the risk framework to scale premarket evidence

31 comments, 54 tags: 7 support, 31 conditional-minor, 15 conditional-major, 1 oppose

Of the 54 points, 7 support FDA's approach as it stands, 31 ask for small changes, 15 for larger ones and 1 opposes it. Most agree that premarket evidence should scale with a device's risk, usually its place on the two axes [1]. Several want FDA to publish a table or matrix from risk position to expected evidence so sponsors can plan [2]; without one, a commenter warns, the approach is "case-by-case review under a new name" [3]. One would scale evidence continuously rather than in rigid bins [4].

Sketched gradients run from benchmarking alone, or with retrospective review, for low-risk informational functions to prospective study for high-risk or autonomous ones [5]. Two warn that requiring trials broadly would favor well-funded incumbents [6], and one, as a larger change, would let benchmarking alone presumptively suffice in the lower and middle bands [7]. Another would instead have action-directing and action-taking devices generally studied prospectively, whatever the consequence level [8].

One would base evidence on residual risk after actual safeguards [9], another wants evidence that assumed safeguards such as clinician review work [10], and a third would ask more where errors are harder for users to detect [11]. As larger changes, one would set evidence by hazard analysis and control effectiveness rather than the axes [12], another by the highest material risk or hard gate rather than an average position [13], and another only after clinical association is established [14]. One wants stricter validation for children and other vulnerable patients [15].

One commenter would have its four-dimension risk profile set the benchmarking elements, test difficulty, confirmation and architectural safeguards [16], illustrating that radiologist review does not spare a high-risk generative report tool the full safeguards [17], and argues that generative implementations need extra scrutiny because reproducibility and bounding must be engineered, though conventional AI is not inherently safe [18]. The opposing point says the framework is not needed, pointing to HIPAA rules instead [19].

**Sources.** [1] 0020 ¶p063; 0025 ¶p329; 0035 ¶p116, ¶p120; 0045 ¶p039; 0049 ¶p044; 0051 ¶p016; 0052 ¶p071; 0056 ¶p032; 0067 ¶p017; 0072 ¶p007; 0087 ¶p050; 0091 ¶p060; 0094 ¶p024; 0100 ¶p040; 0113 ¶p035; 0118 ¶p028; 0123 ¶p003 [2] 0003 ¶p032; 0006 ¶p024; 0013 ¶p054; 0028 ¶p024; 0051 ¶p016; 0093 ¶p017 [3] 0013 ¶p012 [4] 0049 ¶p045 [5] 0006 ¶p024; 0020 ¶p063; 0045 ¶p039; 0113 ¶p035; 0118 ¶p028 [6] 0006 ¶p024; 0094 ¶p024 [7] 0094 ¶p042 [8] 0107 ¶p016 [9] 0120 ¶p164 [10] 0052 ¶p070 [11] 0056 ¶p032 [12] 0055 ¶p107 [13] 0005 ¶p310 [14] 0113 ¶p021 [15] 0047 ¶p003 [16] 0126 ¶p161 [17] 0126 ¶p237 [18] 0126 ¶p246, ¶p248 [19] 0019 ¶p020

## Q9: Benchmarking structure and methods

59 comments, 259 tags: 20 support, 210 conditional-minor, 19 conditional-major, 10 neutral

Of the 259 points, 20 support FDA's approach as it stands, 210 ask for small changes, 19 for larger ones and 10 take no position. Many praise the element set while proposing additions [1]. Several, as a larger change, would have each safety-critical failure, such as a missed escalation or unsafe dosing, meet its own limit, so that strong average performance cannot offset it [2]. Several want results reported by subgroup and setting rather than only in aggregate, naming groups such as children by

developmental stage, non-English speakers, racially and ethnically diverse populations, and rural or community settings [3]; one would judge each group on safe use rather than equal error rates [4].

Proposed additions are many, mostly from single commenters. Several would check sources and patient-specific facts: citation fidelity, faithfulness to sources, invented measurements or history, or claims about the patient checked against the record [5]. Three want devices tested on inputs carrying no signal, treating confident output over absent input as failure [6]. Two would test whether outputs bend to a user's claimed seniority, history or stated beliefs [7], and two whether devices harmfully affirm users, such as by validating a risky decision [8]. Others would test deferral, including over-deferral and unverifiable claims [9], or add elements for clinical needs hidden in administrative requests [10], history completeness against a prespecified question list [11], chronic-disease reasoning and atypical presentations [12], and the strength of evidence behind outputs [13].

On method, three want reproducibility shown over repeated runs on clinical claims or decision fields [14], one noting that zero temperature does not guarantee identical output [15]. Three want behavior over long or multi-turn sessions tested directly, with guardrails holding beyond the first turn or samples drawn late in sessions [16]. Several want human factors, usability or automation bias tested [17]. Three endorse FDA's independence requirement for adjudicators, including LLMs [18]; three would show an LLM judge's independence through a different model family or disclosed lineage [19].

Four want architecture to count: safety rules enforced outside the model, or deterministic designs shown by architectural verification plus targeted testing [20]; one, as a larger change, would require validated stagewise uncertainty, an independent safety-net path and structured deferral for high-risk functions [21]. Several want security testing beyond the current elements, such as prompt injection through retrieved documents, poisoned retrieval and supply-chain attacks [22], and three want, or ask FDA to consider, tests of behavior when dependencies fail or time out [23].

Other larger changes would require a clinical trial with informed consent [24], establish clinical association and acceptable performance before benchmarking [25], have FDA consider a standardized out-of-domain test set [26], or hold benchmarking to bench-test standards, including sequestered datasets the sponsor never sees [27]. On external standards, one suggests standards such as ISO/IEC 42001 and 23894, the NIST AI RMF and CONSORT-AI [28], another asks FDA to survey the field itself [29], and three point to their own open instruments, specifications or benchmarks as examples or starting points [30].

**Sources.** [1] 0003 ¶p033; 0005 ¶p311; 0013 ¶p056; 0020 ¶p015; 0025 ¶p154; 0028 ¶p024; 0043 ¶p018; 0045 ¶p042; 0049 ¶p047; 0052 ¶p081; 0056 ¶p010; 0067 ¶p018; 0078 ¶p004; 0091 ¶p012; 0093 ¶p018; 0094 ¶p044; 0100 ¶p041; 0118 ¶p030; 0120 ¶p169 [2] 0005 ¶p296; 0021 ¶p010; 0025 ¶p159; 0052 ¶p081; 0055 ¶p087; 0056 ¶p014; 0114 ¶p013 [3] 0005 ¶p213; 0017 ¶p053; 0020 ¶p060; 0028 ¶p013; 0051 ¶p017; 0118 ¶p031; 0121 ¶p085; 0126 ¶p292 [4] 0059 ¶p029 [5] 0013 ¶p058; 0054 ¶p075; 0088 ¶p036; 0093 ¶p018; 0111 ¶p024; 0120 ¶p174; 0126 ¶p106 [6] 0062 ¶p031, ¶p048; 0064 ¶p038; 0078 ¶p045 [7] 0058 ¶p025, ¶p029; 0123 ¶p026 [8] 0021 ¶p009; 0098 ¶p034 [9] 0026 ¶p005, ¶p006; 0098 ¶p039; 0126 ¶p236 [10] 0025 ¶p151 [11] 0088 ¶p028 [12] 0118 ¶p030, ¶p032 [13] 0059 ¶p025; 0094 ¶p044 [14] 0076 ¶p054; 0088 ¶p032; 0126 ¶p152 [15] 0126 ¶p205 [16] 0027 ¶p058; 0078 ¶p032; 0088 ¶p034 [17] 0024 ¶p007; 0056 ¶p013; 0067 ¶p023; 0078 ¶p053; 0093 ¶p012; 0094 ¶p045 [18] 0076 ¶p027; 0126 ¶p118; 0127 ¶p022 [19] 0076 ¶p056; 0098 ¶p037; 0123 ¶p052 [20] 0035 ¶p055; 0100 ¶p042; 0111 ¶p021; 0127 ¶p031 [21] 0126 ¶p223, ¶p225, ¶p227 [22] 0013 ¶p057; 0018 ¶p010, ¶p012; 0032 ¶p053, ¶p055; 0078 ¶p051; 0102 ¶p008 [23] 0032 ¶p046; 0098 ¶p030; 0111 ¶p050 [24] 0019 ¶p022 [25] 0113 ¶p023 [26] 0026 ¶p004 [27] 0076 ¶p013 [28] 0093 ¶p018 [29] 0013 ¶p061 [30] 0078 ¶p021; 0085 ¶p030; 0091 ¶p035

## Q10: Benchmark validity and sponsor-developed benchmarks

38 comments, 84 tags: 1 support, 73 conditional-minor, 9 conditional-major, 1 neutral

Many share FDA's concern that public benchmarks may be contaminated by training data or saturated,

so scores can overstate real-world performance [1]. Two propose layers combining public and sequestered benchmarks with clinical confirmation [2].

Several want construct validity "argued, not assumed": sponsors should say what real-world behavior a benchmark predicts and why, and how its cases match the intended use [3]. Several want evidence that scores track an independent reference standard, real-world performance or clinical-confirmation results [4]. Commenters borrow standards from analytical validation [5], clinical laboratory performance specifications [6] and educational testing's five sources of validity evidence [7]. One would qualify the scoring instrument before reading any result: reference answers fixed independently of the adjudicator, automated adjudicators checked against human raters, disagreement rules set in advance and versions recorded [8]. Another wants each gating benchmark to document its reproducibility floor [9]; two suggest documenting the benchmark and its dataset in detail [10].

As sponsors often cannot see a third-party model's training data, one proposes a dated training-cutoff attestation with temporally held-out data [11], another provenance dating and canary items [12]; one asks FDA to accept reasonable demonstrations of independence rather than impossible certainty [13].

Many want some testing on sequestered or held-out sets kept out of development, some only for higher-risk claims or where practical [14]. Proposals include a national repository built with NIST and scored blind by a third party [15], a never-published rotating set kept by FDA or a recognized body, offered as worth evaluating [16], and banks held by professional societies [17].

Many would allow sponsor-built benchmarks, often because no public one fits a narrow or novel use, if safeguarded against tuning to the test [18]. They differ on independence: one cites the conflict of interest in urging review by independent experts not paid according to the outcome [19]; another would require full disclosure instead, since requiring independence would leave small developers no benchmark [20]; a third says a well-documented sponsor benchmark often has higher construct validity than a public one [21].

One warns that tidy, complete prompts overestimate the safety of patient-facing AI [22]; others want clinical friction such as contradictory histories [23], or premise-changing items adjudicated by humans for lay-rescuer functions [24]. One warns that graded outputs may be surrogates whose link to outcomes may not be established [25]; another discounts a device stating its own safety constraints, having seen a system describe a failure and then commit it [26]; a third says items validated on humans need revalidation for models [27]. The one supporting point welcomes FDA raising construct validity alongside independence concerns [28].

**Sources.** [1] 0003 ¶p036; 0005 ¶p226; 0013 ¶p066; 0017 ¶p027; 0027 ¶p073; 0045 ¶p049; 0048 ¶p005; 0067 ¶p024; 0087 ¶p023; 0102 ¶p010; 0114 ¶p016; 0115 ¶p022; 0120 ¶p180 [2] 0045 ¶p049; 0114 ¶p017 [3] 0013 ¶p064; 0017 ¶p027; 0048 ¶p005; 0049 ¶p049; 0115 ¶p022; 0116 ¶p049; 0120 ¶p178 [4] 0028 ¶p024; 0093 ¶p019; 0100 ¶p045; 0113 ¶p024; 0118 ¶p034; 0123 ¶p040 [5] 0013 ¶p064; 0028 ¶p024 [6] 0091 ¶p015 [7] 0123 ¶p033 [8] 0091 ¶p024, ¶p025, ¶p028, ¶p030, ¶p031 [9] 0095 ¶p034 [10] 0035 ¶p060; 0085 ¶p025 [11] 0013 ¶p063 [12] 0067 ¶p025 [13] 0120 ¶p182 [14] 0003 ¶p037; 0005 ¶p217; 0013 ¶p065; 0017 ¶p027; 0035 ¶p060; 0045 ¶p049; 0049 ¶p050; 0051 ¶p018; 0056 ¶p014; 0067 ¶p025; 0088 ¶p016; 0093 ¶p019; 0100 ¶p046; 0102 ¶p010; 0114 ¶p017; 0115 ¶p022; 0116 ¶p048; 0120 ¶p179; 0123 ¶p008 [15] 0003 ¶p037 [16] 0013 ¶p067 [17] 0116 ¶p048; 0118 ¶p034; 0123 ¶p042 [18] 0013 ¶p065; 0028 ¶p024; 0035 ¶p060; 0049 ¶p050; 0051 ¶p018; 0056 ¶p014; 0067 ¶p025; 0082 ¶p031; 0093 ¶p019; 0100 ¶p047; 0118 ¶p034; 0120 ¶p181; 0123 ¶p008 [19] 0113 ¶p024 [20] 0082 ¶p031 [21] 0100 ¶p047 [22] 0025 ¶p172 [23] 0102 ¶p010 [24] 0082 ¶p029 [25] 0107 ¶p012 [26] 0065 ¶p027, ¶p028 [27] 0123 ¶p034 [28] 0091 ¶p013

## Q11: Choosing a clinical confirmation approach

32 comments, 54 tags: 5 support, 44 conditional-minor, 4 conditional-major, 1 oppose

Most back a risk-proportionate menu of confirmation methods and accept that a prospective study is not

always needed [1]. Many link the need for prospective study to action-directing, autonomous, irreversible or high-consequence functions [2]. Several value shadow deployment for showing real-world behavior without exposing patients [3]. Three warn that requiring trials broadly would favor incumbents or price out small firms [4].

Two want FDA to publish presumptive tiers or a matrix keyed to the risk framework [5]; as larger changes, one would have the risk mapping, not sponsor justification, decide [6], and another wants a five-tier ladder climbed in order, the risk determination setting the required tier [7]. Others would scale rigor to the uncertainty left after benchmarking [8], one treating methods as alternatives or complements rather than a fixed hierarchy [9]. One says a lighter approach should rest on a well-qualified benchmark [10].

At the other end, one rejects anything short of a clinical trial [11]. Another wants prospective studies to be the main evidence for devices that directly influence care, since benchmarks may miss how real users describe, read and act on outputs [12], suggesting for example non-inferiority on glycemic control for an insulin-dosing device [13]. One would require confirmation for every device, with retrospective or shadow evidence insufficient at high risk [14]; another sets a floor: action-directing or action-taking functions with severe consequences should never rest on retrospective evaluation alone [15].

On method, one wants submissions to say whether reference judgments were recorded before reviewers saw outputs [16], and adjudicators qualified by task, such as nurse assessors [17]; another wants retrospective tests to give the device only what was known at the decision point [18]. Three want the expected real-world input distribution specified, with confirmation evidence matched to it or labeled by population and setting [19]. For supervised drafting, one would confirm the radiologist-plus-model workflow, such as through aided-versus-unaided reading [20]; others add physical testing of generated physical devices [21], or simulation, cadaveric work and staged introduction for high-risk surgery [22].

**Sources.** [1] 0003 ¶p038; 0005 ¶p314; 0006 ¶p024; 0009 ¶p023; 0013 ¶p068; 0020 ¶p017; 0028 ¶p024; 0030 ¶p008; 0049 ¶p051; 0055 ¶p153; 0087 ¶p024; 0093 ¶p020; 0094 ¶p048; 0100 ¶p040; 0113 ¶p025; 0118 ¶p036; 0120 ¶p186 [2] 0003 ¶p039; 0005 ¶p314; 0006 ¶p024; 0017 ¶p028; 0025 ¶p336; 0049 ¶p052; 0055 ¶p156; 0086 ¶p004; 0093 ¶p020; 0113 ¶p025; 0120 ¶p187 [3] 0003 ¶p038; 0013 ¶p070; 0049 ¶p052; 0118 ¶p036 [4] 0003 ¶p039; 0006 ¶p024; 0094 ¶p048 [5] 0006 ¶p024; 0028 ¶p024 [6] 0013 ¶p069 [7] 0045 ¶p051, ¶p052 [8] 0030 ¶p009; 0052 ¶p085; 0055 ¶p153 [9] 0030 ¶p009 [10] 0091 ¶p061 [11] 0019 ¶p028 [12] 0107 ¶p007, ¶p008, ¶p009, ¶p010 [13] 0107 ¶p014 [14] 0113 ¶p025 [15] 0118 ¶p036 [16] 0009 ¶p027 [17] 0009 ¶p025 [18] 0052 ¶p086 [19] 0005 ¶p255; 0013 ¶p071; 0093 ¶p020 [20] 0120 ¶p160, ¶p161 [21] 0053 ¶p008 [22] 0017 ¶p029

## V-RWD: Real-world and ex-US data in clinical confirmation (no numbered question)

3 comments, 5 tags: 5 conditional-minor

All five points accept such data while asking for small changes [1]. Two want the data's fit to the intended setting shown [2]. One would require a formal transportability assessment of differences in case mix, practice patterns, coding and data capture, with quantitative adjustment such as reweighting where differences are material; ex-US evidence should be "neither privileged nor discounted" [3]. The other supports using such data to enlarge test sets but warns that they leave rare combinations thin, bring population, practice and equipment differences whose relevance must be established, and vary many factors at once, so a drop in performance cannot always be explained, citing pneumonia models that could identify the source hospital [4].

The third would have FDA list large, longitudinal, independently governed clinical-genetic and clinical-outcomes databanks as a confirmation approach between retrospective evaluation and prospective study, if their provenance, consent and adjudication independence are documented, arguing that such real data outperform synthetic data for hard-to-sample groups and avoid its circularity [5].



**Sources.** [1] 0051 ¶p019; 0094 ¶p047; 0126 ¶p261 [2] 0051 ¶p019; 0126 ¶p261 [3] 0051 ¶p019 [4] 0126 ¶p261 [5] 0094 ¶p047

## Q12: Statistical rigor and combining evidence

22 comments, 43 tags: 39 conditional-minor, 2 conditional-major, 2 neutral

Two call the paper's statistical guidance underdeveloped or lacking a path forward [1], one suggesting non-inferiority or Bayesian approaches [2]. Several want prespecified endpoints, units of analysis or sample sizes, with confidence intervals and results by subgroup, site or user type [3]. Three warn that average or prevalence-matched results can miss rare catastrophic errors [4], two wanting separate scenario or challenge evidence for them [5].

Because outputs vary for the same input, several want tests repeated and run-to-run variation reported [6]. One would compare differences against the device's own reproducibility floor rather than zero [7]; another would make the case, not the run, the unit of inference [8], noting that bounding a failure rate below 1% takes about 300 failure-free cases [9]. One commenter wants acceptance judged on the lower confidence bound [10] and sampling settings such as temperature under change control [11]; another wants a stated minimum detectable difference so a null subgroup result is not read as fairness [12].

Several oppose pooling benchmark and confirmation evidence into one estimate, often because the two sample different settings, or allow it only under strict conditions [13]; one fears cheap benchmark data would dominate costly confirmation data [14]. By contrast, one would let small sponsors use a rigorous prospective feasibility study as both [15]. Several likewise favor reporting synthetic and real results separately unless shown compatible [16], with the generation method documented [17].

One wants effect sizes that matter clinically, not just ones that clear a p-value [18]; another warns that rigor cannot rescue a wrong outcome measure such as task completion [19]. One says sample sizes may not be fixable upfront for open-ended outputs, and would first establish clinical association and representativeness [20]; another says claim-level endpoints allow conventional sample-size methods [21].

**Sources.** [1] 0018 ¶p026; 0020 ¶p042 [2] 0018 ¶p027 [3] 0005 ¶p315; 0020 ¶p045; 0052 ¶p088; 0055 ¶p159, ¶p163, ¶p166; 0113 ¶p026; 0120 ¶p189, ¶p190 [4] 0020 ¶p046; 0035 ¶p054; 0052 ¶p087 [5] 0035 ¶p054; 0052 ¶p087 [6] 0013 ¶p073; 0020 ¶p045; 0035 ¶p054; 0055 ¶p162; 0095 ¶p030; 0120 ¶p190; 0123 ¶p044 [7] 0095 ¶p030 [8] 0123 ¶p043 [9] 0123 ¶p046 [10] 0013 ¶p074 [11] 0013 ¶p076 [12] 0095 ¶p038 [13] 0005 ¶p300; 0013 ¶p077; 0051 ¶p020; 0055 ¶p171; 0091 ¶p065; 0093 ¶p021 [14] 0013 ¶p077 [15] 0129 ¶p012 [16] 0005 ¶p315; 0049 ¶p053; 0055 ¶p170; 0120 ¶p192; 0126 ¶p273 [17] 0049 ¶p054; 0093 ¶p021 [18] 0045 ¶p060 [19] 0025 ¶p374 [20] 0113 ¶p026 [21] 0126 ¶p159

## Q13: Suitability and safeguards for synthetic data

26 comments, 52 tags: 3 support, 45 conditional-minor, 2 conditional-major, 2 neutral

Most see synthetic data as a useful supplement, especially for rare events, stress and adversarial tests or controlled variation, but not a replacement for real data [1]. Many share FDA's worry that a generator of the same model class can reproduce the device's blind spots [2]; one calls this worse than no synthetic data, since it manufactures unwarranted confidence [3].

Several want an independent generator or one of different lineage, some only where feasible [4], and nine want the generation method or provenance disclosed [5]. Six want synthetic results checked against real distributions or confirmed on real data [6], or reported separately [7]. One wants labels set or verified apart from the generating model, noting that clinician validation shows a case is coherent, not that the set is representative [8].



Many caution against relying on synthetic data for underrepresented subgroups, often because a generator shares the same gaps [9]. As a larger change, one would bar synthetic data from establishing performance rates [10]. By contrast, one calls validated synthetic data the most direct way to strengthen generalizability evidence, filling combinations of equipment, protocol and population that real test sets leave thin, and wants it accepted in submissions, change plans and re-benchmarking with its validity shown, building on FDA's modeling credibility framework [11].

One warns that synthetic patients are too tidy and wants traits such as minimization, low literacy and proxy questions modeled [12]; another that synthetic cases of rare pediatric presentations may look plausible without matching real ones [13]. Views on fit differ: one finds it suited to common, well-characterized conditions but not rare ones [14], another to inputs with known physics, such as imaging, less so where references depend on outcomes [15], and two find it less reliable for, or no replacement for, real data on workflow, communication or human behavior [16]. For signal simulators, one wants each simulation's purpose and limits stated [17].

**Sources.** [1] 0005 ¶p316; 0013 ¶p081; 0017 ¶p033; 0020 ¶p050; 0025 ¶p339; 0028 ¶p016; 0035 ¶p062; 0045 ¶p062, ¶p063; 0048 ¶p006; 0049 ¶p055; 0052 ¶p089; 0055 ¶p172, ¶p179; 0076 ¶p055; 0091 ¶p063; 0093 ¶p022; 0113 ¶p027; 0115 ¶p023; 0120 ¶p194, ¶p195; 0126 ¶p276 [2] 0003 ¶p041; 0005 ¶p260; 0013 ¶p079; 0017 ¶p033; 0020 ¶p048; 0028 ¶p016; 0045 ¶p063; 0049 ¶p056; 0055 ¶p186; 0076 ¶p055; 0093 ¶p022; 0113 ¶p027; 0118 ¶p040; 0120 ¶p196 [3] 0013 ¶p079 [4] 0005 ¶p149; 0045 ¶p063; 0049 ¶p056; 0055 ¶p186; 0076 ¶p055; 0093 ¶p022; 0126 ¶p274 [5] 0013 ¶p084; 0017 ¶p033; 0028 ¶p016; 0049 ¶p019; 0085 ¶p038; 0093 ¶p022; 0113 ¶p024; 0115 ¶p023; 0120 ¶p197 [6] 0005 ¶p244; 0028 ¶p016; 0049 ¶p056; 0055 ¶p187; 0093 ¶p022; 0120 ¶p197 [7] 0017 ¶p033; 0048 ¶p006; 0115 ¶p023 [8] 0091 ¶p064, ¶p066 [9] 0003 ¶p042; 0013 ¶p083; 0048 ¶p006; 0051 ¶p001; 0055 ¶p184; 0076 ¶p055; 0093 ¶p022; 0113 ¶p027; 0115 ¶p023; 0118 ¶p040 [10] 0013 ¶p082 [11] 0126 ¶p044, ¶p258, ¶p272, ¶p293 [12] 0025 ¶p340, ¶p346, ¶p347 [13] 0020 ¶p049 [14] 0118 ¶p040 [15] 0126 ¶p271 [16] 0049 ¶p055; 0055 ¶p179, ¶p180, ¶p182 [17] 0074 ¶p022

## Q14: Comparators and acceptance criteria for open-ended outputs

33 comments, 108 tags: 2 support, 89 conditional-minor, 12 conditional-major, 1 oppose, 4 neutral

Most want human-AI team performance as the basis, or a principal basis, of evaluation when a clinician reviews the output [1]; six would make device-alone performance the basis for autonomous use [2]. Several still want device-alone results reported, as supporting evidence or to characterize failure modes [3]; one adds that a reviewer may not raise a strong model's accuracy, so the human's role is accountability [4]. Nine say oversight must be tested, not presumed, by measuring automation bias, detection or override rates, or how often clinicians adopt wrong outputs [5]. One wants errors prevented counted alongside errors introduced [6]. Another wants the output's place in the workflow, and whether an independent clinician assessment precedes it, fixed in the intended use, arguing that the same clinicians will perform differently under different orderings and citing a study in which ordering alone changed patients' trust [7]; one warns against testing as a team product while marketing autonomous use [8]. Proposed designs compare intended users with and without the device [9], or add device-alone and specialist-panel arms [10].

Commenters divide on the standard. Several would not let median practice set the bar, at least for safety, often because it is unmeasured or normalizes gaps in care [11]; one, as a larger change, urges a common, nationally recognized evidence-based reference framework [12]. Two favor the median clinician: one as default, warning that a consensus panel sets a bar practicing clinicians do not meet [13], the other where no ground truth exists [14]; one would use the median specialist where specialists are unavailable, to fix an absolute threshold [15]. Two want the standard justified per intended use rather than fixed globally [16].

On generalists and specialists, some would match the comparator to the labeled user or the clinician

who actually does the task [17]; others want specialists to set the reference for specialty tasks [18], three even when generalists use the device [19].

Several want the reference panel's internal agreement measured and reported [20], two calling the panel a measurement instrument [21]; one says no stable human reference exists for Medicaid functional assessment, so sponsors should demonstrate one [22], and one wants sets of acceptable answers so legitimate disagreement is not scored as error [23]. Several want acceptance criteria prespecified [24], covering several dimensions or graded by consequence [25]. Two want safety criteria stricter than for any one clinician, since a device repeats its errors at scale [26]. As larger changes, one wants conjunctive safety standards that proficiency elsewhere cannot offset [27], another deliberately falsifiable thresholds at half and all of a specialty clinician baseline for supervised and independent authorization [28], and another would score the discrete clinical claims in open-ended output against a reference standard [29]. The opposing point doubts physicians would join panels, fearing backlash and risk to their licenses [30].

**Sources.** [1] 0013 ¶p094; 0017 ¶p030; 0028 ¶p017; 0035 ¶p062; 0046 ¶p018; 0052 ¶p094; 0054 ¶p076; 0055 ¶p192; 0056 ¶p017; 0070 ¶p009; 0086 ¶p005; 0087 ¶p027; 0093 ¶p023; 0094 ¶p050; 0100 ¶p051; 0118 ¶p042; 0120 ¶p202; 0123 ¶p053; 0126 ¶p176 [2] 0035 ¶p062; 0093 ¶p023; 0094 ¶p050; 0100 ¶p051; 0118 ¶p042; 0126 ¶p177 [3] 0013 ¶p094; 0028 ¶p024; 0046 ¶p018; 0055 ¶p192; 0056 ¶p017; 0120 ¶p203; 0123 ¶p053 [4] 0046 ¶p018 [5] 0013 ¶p095; 0017 ¶p031; 0054 ¶p006; 0056 ¶p017; 0058 ¶p036; 0076 ¶p058; 0086 ¶p005; 0094 ¶p051; 0123 ¶p053 [6] 0120 ¶p090, ¶p092 [7] 0070 ¶p012, ¶p013 [8] 0094 ¶p051 [9] 0052 ¶p094; 0087 ¶p027 [10] 0056 ¶p016 [11] 0003 ¶p043; 0013 ¶p088; 0025 ¶p184; 0055 ¶p191; 0056 ¶p015; 0076 ¶p058; 0123 ¶p050 [12] 0025 ¶p085 [13] 0100 ¶p049 [14] 0113 ¶p029 [15] 0126 ¶p180, ¶p181 [16] 0028 ¶p024; 0093 ¶p023 [17] 0006 ¶p025; 0013 ¶p093; 0045 ¶p029; 0049 ¶p057 [18] 0055 ¶p190; 0056 ¶p015; 0118 ¶p043 [19] 0056 ¶p009; 0076 ¶p059; 0123 ¶p050 [20] 0009 ¶p033; 0013 ¶p092; 0043 ¶p023; 0100 ¶p049; 0106 ¶p013; 0123 ¶p049 [21] 0013 ¶p092; 0123 ¶p049 [22] 0009 ¶p032 [23] 0123 ¶p049 [24] 0013 ¶p090; 0056 ¶p018; 0076 ¶p058; 0113 ¶p029; 0126 ¶p033 [25] 0049 ¶p058; 0055 ¶p189; 0056 ¶p018; 0120 ¶p201 [26] 0076 ¶p049; 0123 ¶p051 [27] 0123 ¶p051 [28] 0045 ¶p110, ¶p111 [29] 0126 ¶p029 [30] 0019 ¶p034

## Q15: Comparing with care in the device's absence

25 comments, 45 tags: 7 support, 28 conditional-minor, 4 conditional-major, 3 oppose, 3 neutral

Most accept comparing with the care likely without the device, at least for some devices or analyses [1]. Many note that for access-expanding devices the realistic alternative is often no care, delayed care or less qualified help, so a device can benefit patients without matching a specialist [2]. Realistic alternatives named include what patients find searching online [3], an unaided bystander [4], a rushed paper assessment form [5], and unaided radiologists [6].

Several want the counterfactual documented with data rather than assumed, such as from care-access or service records, record review or target-trial emulation [7]; one wants the comparator carried into labeling to prevent unevaluated scope creep [8]. Several also want absolute safety assessed, so that "better than today" is not mistaken for acceptably safe [9], and one would use it only to supplement an available specialist panel [10]; as larger changes, one wants an absolute floor for high-risk functions [11], another absolute, per-condition acceptance thresholds, leaving the counterfactual to the benefit-risk decision [12]. Two warn against choosing a weak comparator [13].

For communication aids used by nonverbal people, one commenter calls the device's absence the only realistic comparator [14] and says a fabricating device scores below it, citing a recording in which 29 percent of transcribed words came from audio with no signal [15]; as a larger change, it wants the absence comparator used as a pass-fail gate alongside the others, required where users cannot check a durable record only the device could produce [16].

Of the three opposing points, two from one commenter reject comparing autonomous functions with no or delayed interpretation, since an unavailable alternative changes a device's benefit, not the harm of its

errors [17]; the third rejects unaided judgment and delayed review as comparators and questions who could justify a machine's choice of no intervention [18].

**Sources.** [1] 0003 ¶p044; 0005 ¶p318; 0006 ¶p025; 0009 ¶p035; 0013 ¶p096; 0017 ¶p032; 0020 ¶p041; 0025 ¶p202; 0045 ¶p066; 0049 ¶p060; 0051 ¶p021; 0052 ¶p092; 0055 ¶p193; 0064 ¶p009; 0082 ¶p032; 0093 ¶p024; 0100 ¶p050; 0113 ¶p030; 0118 ¶p046; 0120 ¶p205; 0123 ¶p050; 0126 ¶p182 [2] 0003 ¶p044; 0006 ¶p025; 0013 ¶p096; 0020 ¶p040; 0045 ¶p066; 0049 ¶p060; 0051 ¶p021; 0100 ¶p050; 0118 ¶p046; 0120 ¶p208 [3] 0006 ¶p025; 0100 ¶p050 [4] 0082 ¶p032 [5] 0009 ¶p036 [6] 0120 ¶p087, ¶p206 [7] 0006 ¶p025; 0009 ¶p040; 0013 ¶p097; 0051 ¶p021; 0093 ¶p024; 0100 ¶p050; 0120 ¶p209 [8] 0013 ¶p097 [9] 0017 ¶p032; 0021 ¶p011; 0025 ¶p203, ¶p205; 0123 ¶p050; 0126 ¶p182 [10] 0118 ¶p046 [11] 0017 ¶p032 [12] 0126 ¶p182 [13] 0049 ¶p061; 0064 ¶p028 [14] 0064 ¶p009 [15] 0064 ¶p011, ¶p012 [16] 0064 ¶p030, ¶p035 [17] 0126 ¶p033, ¶p180 [18] 0019 ¶p036

## Q16: Role of independent third parties

32 comments, 65 tags: 7 support, 49 conditional-minor, 8 conditional-major, 1 oppose

Most see roles for independent third parties, chiefly holding sequestered test sets, running benchmarks and adjudicating results [1]. As larger changes, four would require independent testing or adjudication, or call it necessary, at least for higher-risk devices or safety-critical elements [2]; for lower-risk functions, one would accept independence through uninvolved radiologists or blinded review [3].

Many warn that third parties must not become a bottleneck or toll gate for small sponsors, proposing several accredited bodies on published criteria, often on the ASCA model, scaled or capped fees, and no exclusivity [4]. Several want it kept optional, with sponsors free to submit their own evidence [5]; the opposing point fears third parties would game the system and draw out testing for profit [6].

On independence, several want conflicts or financial ties disclosed, or pay not tied to results [7], and several independence from the foundation model developer as well as the sponsor [8], two adding that an LLM adjudicator should come from a different model family [9]. Two want adjudicators or assessors with matching clinical expertise [10]. One would bar bodies that advised a sponsor from assessing its device [11]; another would define independence relative to the device, so device developers can also build evaluation tools [12].

Several stress that the sponsor stays accountable or that FDA keeps the decision [13]. Three prefer qualifying evaluation assets, or open, reproducible methods with targeted adjudication, to certifying devices [14]; one would base trust on how an evaluation is run: prespecified, automatically scored, reported identically to FDA and sponsor, and auditable [15]. Three see a role for professional societies [16], such as holding specialty datasets or supplying adjudicators [17], and one wants third-party assessments published or available on request [18]. As a larger change, one wants CLIA laboratory accreditation and proficiency testing accepted as meeting some or all of these elements for functions embedded in laboratory testing [19]. One knows of no third-party assessment capacity for generative devices at scale [20].

**Sources.** [1] 0003 ¶p045; 0005 ¶p319; 0006 ¶p028; 0013 ¶p098; 0017 ¶p034; 0025 ¶p210; 0028 ¶p024; 0029 ¶p015; 0035 ¶p064; 0045 ¶p069; 0049 ¶p062; 0055 ¶p195; 0056 ¶p019; 0076 ¶p057; 0091 ¶p067; 0093 ¶p025; 0100 ¶p052; 0115 ¶p024; 0118 ¶p048; 0120 ¶p214; 0123 ¶p042; 0126 ¶p279 [2] 0047 ¶p005; 0056 ¶p005; 0071 ¶p012; 0113 ¶p031 [3] 0120 ¶p215 [4] 0003 ¶p046; 0005 ¶p319; 0006 ¶p028; 0013 ¶p100; 0028 ¶p024; 0035 ¶p064; 0045 ¶p069; 0049 ¶p063; 0055 ¶p203; 0071 ¶p015; 0076 ¶p057; 0093 ¶p025; 0094 ¶p025; 0100 ¶p053; 0118 ¶p048; 0120 ¶p216 [5] 0013 ¶p103; 0094 ¶p025; 0100 ¶p054; 0120 ¶p216; 0126 ¶p291 [6] 0043 ¶p025 [7] 0013 ¶p102; 0025 ¶p211; 0035 ¶p064; 0055 ¶p202; 0071 ¶p014; 0115 ¶p024; 0118 ¶p048 [8] 0067 ¶p028; 0071 ¶p014; 0093 ¶p025; 0100 ¶p055; 0113 ¶p031 [9] 0071 ¶p014; 0100 ¶p055 [10] 0055 ¶p201; 0056 ¶p020 [11] 0013 ¶p101 [12] 0126 ¶p284 [13] 0049 ¶p063; 0055 ¶p204; 0093 ¶p025; 0116 ¶p050; 0120 ¶p217; 0126 ¶p291 [14] 0067 ¶p028; 0091 ¶p067; 0126 ¶p291 [15] 0126 ¶p285, ¶p286, ¶p289 [16] 0116 ¶p050; 0118 ¶p048; 0123 ¶p042 [17] 0118 ¶p048; 0123 ¶p042 [18] 0118 ¶p048 [19] 0094 ¶p054 [20] 0013 ¶p104

## Q17: Other model architectures

16 comments, 26 tags: 2 support, 21 conditional-minor, 3 neutral

Most say the competency-based approach can apply across architectures, often judging the device's clinical behavior rather than its model type [1]; one warns that architecture-specific rules would be obsolete before they were finalized [2].

Most would still add architecture-specific tests [3]. For multimodal devices these include cross-modal consistency or grounding [4], missing, conflicting or mismatched inputs [5], and variation in image acquisition, quality or skin tone [6]. One reports an experiment in which models asked about an absent image still gave diagnoses that shifted with a demographic phrase, perhaps reflecting prevalence, and in one condition mostly filled a structured diagnosis field while saying in text that no image was present; it wants abstention measured in every output channel and models' own reports of input adequacy distrusted [7]. For world models, one suggests testing error over predicted steps and counterfactual results [8], another drawing on FDA's credibility practices for computational modeling [9].

One warns that benchmarks for these architectures are thinner, so clinical confirmation should carry more weight [10]; another that conversational elements do not fit imaging functions that return measurements or segmentations [11]; a third wants a branching point between content-generating and world-model devices before benchmark elements are chosen [12]. One wants companion guidance for multimodal devices, with each modality and the combined device benchmarked [13], another applicability notes for each architecture [14].

**Sources.** [1] 0005 ¶p321; 0025 ¶p376; 0045 ¶p072; 0049 ¶p064; 0055 ¶p108; 0093 ¶p026; 0108 ¶p001; 0113 ¶p033; 0118 ¶p050; 0120 ¶p219; 0123 ¶p054 [2] 0045 ¶p072 [3] 0003 ¶p047; 0005 ¶p321; 0049 ¶p064; 0055 ¶p109; 0093 ¶p026; 0108 ¶p010; 0113 ¶p033; 0118 ¶p050; 0120 ¶p220; 0123 ¶p055 [4] 0005 ¶p321; 0049 ¶p064; 0055 ¶p109; 0113 ¶p033; 0120 ¶p220 [5] 0055 ¶p110; 0108 ¶p003; 0123 ¶p055 [6] 0055 ¶p113; 0093 ¶p026; 0118 ¶p050; 0123 ¶p057 [7] 0108 ¶p003, ¶p005, ¶p006, ¶p007, ¶p010 [8] 0055 ¶p114, ¶p115 [9] 0093 ¶p026 [10] 0013 ¶p105, ¶p106 [11] 0126 ¶p167, ¶p168 [12] 0129 ¶p014 [13] 0003 ¶p047 [14] 0093 ¶p026

## POSTMARKET: Oversight once a device is in use

91 comments, 796 tags: 30 support, 697 conditional-minor, 31 conditional-major, 12 oppose, 26 neutral

### Q18: Trading premarket uncertainty for postmarket monitoring

43 comments, 112 tags: 2 support, 88 conditional-minor, 7 conditional-major, 12 oppose, 3 neutral

Most accept trading some premarket uncertainty for stronger monitoring, but only within limits and under conditions [1]. Supporters argue that no premarket test can cover open-ended inputs, and that monitoring tracks devices that change after deployment, faster and at lower cost [2]; one sees supervised report drafting, whose output a specialist intercepts, as a strong candidate [3]; as a larger change, one wants the trade to be the default outside the highest-risk band [4].

Five commenters oppose substitution, at least outside the lowest-risk functions: monitoring should verify or supplement premarket evidence, not replace it [5]. Reduced evidence, one argues, shifts the discovery of failures to patients, and the conditions that could justify it rarely hold together for high-risk functions [6]; another would allow it only for low-consequence, non-directive functions, and only once a GenAI reporting taxonomy, public thresholds and a patient reporting channel exist [7].

Most would not extend the trade to autonomous, high-consequence, irreversible or time-critical functions, or wherever harm could come before monitoring detects it [8]. Further exclusions proposed cover errors users cannot detect [9], users who cannot report failures [10], devices working offline [11], ground truth that is delayed or unavailable [12], permanent implants [13], and failures that arise within a session and reset before any periodic check [14]. Three would make irreversibility a gate of its own [15], one regardless of severity [16].

Where the trade is allowed, many want a prespecified, enforceable monitoring plan with thresholds, triggers and consequences such as restriction, suspension or rollback [17]. Several want detection capability demonstrated rather than assumed: what the program covers and misses, how fast it detects, and whether it catches the failure modes premarket evidence left open [18]. Others want monitoring running from the first day of deployment [19], records independent of the device or tamper-evident [20], a machine supervisor's reliability shown first [21], near real-time monitoring of safety-critical behavior [22], or, for users who cannot check outputs, a named caregiver or clinician notified of faults within the window of harm [23]; as larger changes, one wants automatic runtime circuit breakers [24], another deploying sites able to inventory, set permissions for, log and stop the device [25].

Skeptics draw on experience: one, noting that postmarket obligations are generally met less reliably, wants FDA to publish completion data first [26]; another, from drug safety, that surveillance finds mainly what it was designed to find and tends to count timeliness rather than detection [27]; a third that zero adverse events may mean blind monitoring [28]; and a fourth that GenAI failures lack a reporting taxonomy or a natural reporter [29].

**Sources.** [1] 0003 ¶p049; 0005 ¶p322; 0006 ¶p026; 0013 ¶p109; 0017 ¶p036; 0020 ¶p054; 0025 ¶p213; 0028 ¶p025; 0035 ¶p065; 0045 ¶p074; 0048 ¶p007; 0049 ¶p067; 0052 ¶p111; 0054 ¶p079; 0055 ¶p205; 0075 ¶p019; 0077 ¶p017; 0090 ¶p009; 0093 ¶p027; 0099 ¶p023; 0100 ¶p057; 0102 ¶p006; 0105 ¶p026; 0115 ¶p025; 0116 ¶p011; 0118 ¶p052; 0120 ¶p226; 0127 ¶p039; 0130 ¶p004 [2] 0003 ¶p049; 0006 ¶p026; 0094 ¶p059, ¶p060; 0100 ¶p057 [3] 0120 ¶p114 [4] 0094 ¶p057 [5] 0023 ¶p006; 0076 ¶p036; 0086 ¶p008; 0113 ¶p036; 0126 ¶p187 [6] 0126 ¶p187, ¶p194 [7] 0076 ¶p036 [8] 0005 ¶p322; 0013 ¶p113; 0017 ¶p036; 0020 ¶p055; 0025 ¶p218; 0028 ¶p025; 0035 ¶p066; 0045 ¶p075; 0048 ¶p007; 0049 ¶p068; 0051 ¶p022; 0052 ¶p118; 0053 ¶p009; 0054 ¶p080; 0055 ¶p214; 0075 ¶p021; 0090 ¶p015; 0093 ¶p027; 0094 ¶p061; 0100 ¶p057; 0105 ¶p026; 0115 ¶p025; 0116 ¶p025; 0118 ¶p052; 0120 ¶p228; 0126 ¶p194; 0131 ¶p003 [9] 0051 ¶p022 [10] 0077 ¶p043; 0093 ¶p027 [11] 0077 ¶p048; 0093 ¶p027 [12] 0090 ¶p015 [13] 0053 ¶p009 [14] 0077 ¶p019 [15] 0013 ¶p114; 0077 ¶p050; 0131 ¶p003 [16] 0131 ¶p027 [17] 0003 ¶p050; 0006

¶p026; 0013 ¶p111; 0028 ¶p025; 0035 ¶p066; 0045 ¶p074; 0055 ¶p210, ¶p212; 0075 ¶p020; 0090 ¶p014; 0093 ¶p027; 0094 ¶p061; 0099 ¶p025; 0100 ¶p057; 0120 ¶p116 [18] 0008 ¶p013; 0013 ¶p112; 0033 ¶p014; 0077 ¶p031; 0093 ¶p027; 0099 ¶p024 [19] 0090 ¶p011; 0116 ¶p024 [20] 0077 ¶p035; 0090 ¶p013; 0127 ¶p039 [21] 0111 ¶p038 [22] 0118 ¶p052 [23] 0077 ¶p040 [24] 0102 ¶p006 [25] 0130 ¶p004, ¶p009 [26] 0013 ¶p110 [27] 0033 ¶p010, ¶p012 [28] 0008 ¶p012 [29] 0076 ¶p036

## Q19: Postmarket performance evaluation approaches

72 comments, 257 tags: 13 support, 226 conditional-minor, 6 conditional-major, 12 neutral

Many accept FDA's three approaches, periodic re-benchmarking, sampled clinician review and degradation monitoring; 226 of the 257 points ask for small changes [1]. Several call the three complementary rather than alternatives [2]. One reports running such clinician review at scale for years [3].

Many want reassessment both on a schedule and on triggering events [4]. Named triggers include model, prompt, retrieval, guardrail or tool changes, including third-party model updates; input shifts or new sites; adverse-event signals; subgroup decline; and breached thresholds [5]. Three see the schedule as a floor for what no trigger catches, such as slow or undeclared drift, or a monitor that never alarms [6]. Two tier cadence by risk, from annual review for low-risk devices to quarterly reviews or monthly drift checks for high-risk ones [7]. Rarer triggers include a new EHR version at the deployer [8], peer regulators' actions [9] and rising adjudicator disagreement [10].

Many want monitoring to reach past model accuracy to clinical consequences, such as delayed care, missed escalation, near misses, overrides, subgroup performance and automation bias [11]. Three would pool or link data across organizations [12]; as a larger change, one wants participation in a qualified postmarket data system required for action-directing or higher-consequence functions [13]. Others would watch for oversight drift, such as checkpoints approved reflexively [14], and for benchmarks that stop being valid or independent [15].

Several want at least one check independent of the device's, or its maker's, own account [16]. One commenter reports audio capture losing a growing share of speech within one session, upstream of the model and with no log entry, and wants missing input logged like errors [17]; a related submission from the same organization wants degradation measured within sessions, since each re-benchmark starts a fresh one [18], and detected fabrications reported to a named caregiver or clinician by session end for users who cannot check outputs themselves [19].

On clinician review, several want random samples combined with, or enriched for, high-consequence, high-acuity or unusual cases [20]. One wants criteria with decision rules, measured adjudicator agreement and recalibration, so reviewers do not diverge unseen [21]; another wants adversarial re-benchmarking on its own cadence, since representative samples are not designed to test resistance to manipulation [22]; a third would reserve clinician review for higher-risk devices or performance concerns [23].

Several want each signal tied to prespecified thresholds and actions such as investigation, restriction, rollback or suspension [24], one adding a human-operated kill switch [25]; as a larger change, another wants automatic lockout and 24-hour reporting for high-materiality events [26]. Several want records that can reconstruct an event, including the model and configuration versions, prompts, tools and human checkpoints involved [27]; some would limit what is kept, through metadata, hashes or risk-tiered retention, to protect privacy [28]. One notes most institutions cannot record which configuration produced an output [29]. Two ask FDA to clarify when a plausible but wrong output is a reportable malfunction [30]; one would report caught errors as aggregate rates [31].



Others ask for proportion: intensity matched to why premarket evidence is incomplete, with no device exempt [32]; re-benchmarking on change, not drift monitoring, for frozen on-device models [33]; and automated metrics accepted for small developers [34].

**Sources.** [1] 0013 ¶p116; 0020 ¶p018; 0028 ¶p019; 0029 ¶p020; 0049 ¶p069; 0052 ¶p121; 0058 ¶p039; 0063 ¶p009; 0067 ¶p029; 0071 ¶p017; 0087 ¶p029; 0090 ¶p021; 0093 ¶p028; 0100 ¶p058; 0115 ¶p026; 0116 ¶p043; 0118 ¶p054; 0120 ¶p230; 0126 ¶p196 [2] 0013 ¶p116; 0052 ¶p121; 0071 ¶p017; 0090 ¶p021; 0093 ¶p028; 0118 ¶p054 [3] 0100 ¶p058 [4] 0003 ¶p053; 0005 ¶p323; 0013 ¶p116; 0017 ¶p039; 0035 ¶p073; 0045 ¶p078; 0049 ¶p070; 0052 ¶p123; 0054 ¶p081; 0056 ¶p022; 0058 ¶p041; 0075 ¶p022; 0090 ¶p022; 0091 ¶p058; 0093 ¶p028; 0100 ¶p060; 0105 ¶p027; 0115 ¶p026; 0116 ¶p044; 0118 ¶p054; 0120 ¶p233, ¶p234 [5] 0013 ¶p117; 0035 ¶p074; 0049 ¶p070; 0054 ¶p092; 0056 ¶p023; 0105 ¶p028; 0115 ¶p027, ¶p028, ¶p029; 0120 ¶p234 [6] 0013 ¶p117; 0090 ¶p022; 0091 ¶p058 [7] 0003 ¶p052; 0113 ¶p037 [8] 0130 ¶p010 [9] 0003 ¶p053 [10] 0033 ¶p028 [11] 0017 ¶p038; 0021 ¶p018; 0025 ¶p222, ¶p228; 0039 ¶p004; 0047 ¶p008; 0049 ¶p106; 0051 ¶p023; 0052 ¶p121; 0054 ¶p091; 0055 ¶p223, ¶p224; 0056 ¶p024; 0086 ¶p009; 0087 ¶p029; 0116 ¶p032; 0120 ¶p231; 0121 ¶p014; 0122 ¶p020 [12] 0021 ¶p018; 0051 ¶p023; 0116 ¶p013 [13] 0116 ¶p022 [14] 0042 ¶p058; 0052 ¶p129 [15] 0052 ¶p109; 0104 ¶p016, ¶p019 [16] 0008 ¶p015; 0062 ¶p044; 0063 ¶p028; 0078 ¶p088; 0093 ¶p028; 0103 ¶p015; 0127 ¶p039 [17] 0062 ¶p022, ¶p026, ¶p033, ¶p046 [18] 0063 ¶p014, ¶p019 [19] 0063 ¶p041 [20] 0055 ¶p225; 0056 ¶p023; 0075 ¶p024; 0092 ¶p062; 0100 ¶p059; 0116 ¶p043; 0118 ¶p054 [21] 0033 ¶p019, ¶p022, ¶p024, ¶p026 [22] 0027 ¶p006 [23] 0028 ¶p019 [24] 0005 ¶p290, ¶p291; 0024 ¶p009; 0047 ¶p009; 0048 ¶p008; 0052 ¶p110; 0055 ¶p226; 0067 ¶p030; 0075 ¶p022; 0105 ¶p028; 0116 ¶p044; 0118 ¶p055; 0122 ¶p023 [25] 0118 ¶p058 [26] 0102 ¶p010 [27] 0035 ¶p113, ¶p114; 0047 ¶p006; 0049 ¶p071; 0052 ¶p141; 0053 ¶p013; 0087 ¶p043; 0092 ¶p015; 0098 ¶p043; 0102 ¶p008; 0115 ¶p032; 0116 ¶p031; 0121 ¶p049; 0130 ¶p010 [28] 0035 ¶p072; 0049 ¶p074; 0092 ¶p087, ¶p088; 0101 ¶p008; 0116 ¶p036 [29] 0050 ¶p009 [30] 0013 ¶p128; 0033 ¶p037, ¶p038 [31] 0013 ¶p129 [32] 0126 ¶p204, ¶p208 [33] 0082 ¶p033 [34] 0111 ¶p040

## Q20: Machine-based supervisory agents

42 comments, 102 tags: 5 support, 90 conditional-minor, 1 conditional-major, 6 neutral

Most accept supervisory agents as an aid to monitoring, but only if the agent is itself evaluated; of the 102 points, 90 ask for small changes and five support the idea as it stands [1]. Two see automated supervision as probably the only option at high volumes [2]; one notes a supervisory layer already used by an EU-certified breast-screening device [3].

Most want the supervisor independent of what it watches, warning that one sharing the device's model, data, dependencies or maker may share its blind spots [4]. Three would use a different model family or company [5]; two caution that a different vendor or model does not by itself make it independent or reliable [6].

Many want people to keep the final say, with the agent triaging or flagging and qualified humans adjudicating [7]. Three want humans to sample cases the agent cleared, not only those it flagged [8]; one cautions that automation suits deterministic checks rather than clinical judgment, and that throughput should not be reported as assurance [9].

Validation requests include sensitivity to specific harms rather than overall agreement [10]; misses and false alarms assessed separately [11]; comparison with human adjudication or outcomes, not another model's agreement [12]; repeated checks over time [13]; and revalidation when the supervisor, including its prompt, changes [14]. Four would treat the supervisor as a safety component or device function in its own right [15]; one suggests qualifying monitoring methods through FDA's tool-qualification program [16].

Others would bound the agent's authority: separating observing, recommending and intervening [17], running observation-only before it may alter output [18], limiting permissions with a safe or fail-closed default [19], and keeping fixed safety limits in deterministic rules [20]. As a larger change, one wants supervision configured by the deploying institution where the device acts, able to deny out-of-scope actions [21].

Experience shapes some views. Related submissions from one organization report an LLM reviewer that confidently inverted cause and effect, attributed an unspoken sentence to a speaker and answered

from a three-day-old copy, each caught only because a person knew better [22]; one of them wants agents validated on cases no operator can check, findings that state their source and its age, and stale sources treated as failures [23]. One developer's verifier would have flagged a true statement as fabricated because its ground truth was drawn too narrowly [24]; another commenter cites judge models whose verdicts shifted with rephrased prompts [25].

On records, one wants agent output that cannot be mistaken for human judgment [26]; another notes the current HL7 audit standard cannot record one agent acting for another [27]; three want append-only, immutable or tamper-evident logs [28]. Four stress that an agent cannot carry accountability, which must stay clearly with the manufacturer, deployer or people overseeing it [29].

**Sources.** [1] 0003 ¶p055; 0005 ¶p324; 0013 ¶p123; 0017 ¶p040; 0025 ¶p240; 0028 ¶p019; 0033 ¶p030; 0035 ¶p075; 0045 ¶p081; 0049 ¶p075; 0052 ¶p130; 0055 ¶p229; 0056 ¶p030; 0062 ¶p037; 0066 ¶p008, ¶p047; 0071 ¶p024; 0075 ¶p042; 0076 ¶p063; 0090 ¶p025; 0093 ¶p029; 0095 ¶p044; 0100 ¶p062; 0102 ¶p010; 0105 ¶p029; 0106 ¶p017; 0111 ¶p032; 0113 ¶p038; 0115 ¶p034; 0116 ¶p047; 0120 ¶p238; 0126 ¶p211; 0127 ¶p040 [2] 0013 ¶p119; 0090 ¶p024 [3] 0126 ¶p210 [4] 0005 ¶p324; 0013 ¶p120; 0017 ¶p040; 0032 ¶p101; 0035 ¶p076; 0052 ¶p131; 0055 ¶p237; 0056 ¶p030; 0062 ¶p037; 0066 ¶p051; 0071 ¶p026; 0075 ¶p044; 0076 ¶p063; 0090 ¶p026; 0093 ¶p029; 0095 ¶p044; 0105 ¶p029; 0106 ¶p017; 0110 ¶p003; 0115 ¶p035; 0116 ¶p080; 0118 ¶p057; 0120 ¶p239; 0130 ¶p011 [5] 0013 ¶p122; 0055 ¶p237; 0110 ¶p003 [6] 0075 ¶p044; 0115 ¶p034 [7] 0003 ¶p055; 0005 ¶p324; 0013 ¶p124; 0025 ¶p237; 0033 ¶p030; 0052 ¶p132; 0076 ¶p063; 0086 ¶p009; 0093 ¶p029; 0100 ¶p062; 0102 ¶p010; 0105 ¶p030; 0113 ¶p038; 0116 ¶p047; 0118 ¶p057; 0120 ¶p240 [8] 0055 ¶p239; 0075 ¶p045; 0093 ¶p029 [9] 0033 ¶p031 [10] 0013 ¶p123 [11] 0033 ¶p030; 0075 ¶p044; 0111 ¶p035 [12] 0076 ¶p063; 0090 ¶p025; 0116 ¶p047 [13] 0013 ¶p123; 0090 ¶p025 [14] 0013 ¶p125; 0090 ¶p028; 0095 ¶p044; 0105 ¶p030 [15] 0049 ¶p080; 0055 ¶p229; 0115 ¶p041; 0126 ¶p211 [16] 0090 ¶p029 [17] 0049 ¶p077; 0075 ¶p042 [18] 0111 ¶p036 [19] 0035 ¶p076; 0105 ¶p030 [20] 0055 ¶p238; 0071 ¶p025; 0115 ¶p036 [21] 0130 ¶p011 [22] 0062 ¶p035, ¶p036; 0066 ¶p009, ¶p025 [23] 0066 ¶p047, ¶p049, ¶p050 [24] 0111 ¶p033 [25] 0095 ¶p043 [26] 0131 ¶p049, ¶p051 [27] 0042 ¶p036, ¶p041 [28] 0090 ¶p027; 0105 ¶p030; 0115 ¶p039 [29] 0025 ¶p241; 0033 ¶p029; 0049 ¶p076; 0102 ¶p010

## Q21: Shared ecosystem roles and accountability

42 comments, 112 tags: 7 support, 99 conditional-minor, 3 conditional-major, 3 neutral

Most accept shared roles on one condition: others supply signals, data and expertise, while the manufacturer keeps responsibility for investigating, acting and reporting; of the 112 points, 99 ask for small changes and seven support FDA's position as it stands [1]. One puts it as information coming from anywhere while "the duty to investigate and act remains with the manufacturer" [2]; another, that outsourcing transfers work, not accountability [3]; a third, that contracts can allocate tasks but not accountability [4].

Many want each party's duties spelled out, often in proportion to what it controls, so no one can later claim a failure was someone else's [5]. Proposed splits give manufacturers design, validation, monitoring and change control; deploying institutions local integration, training, access and incident handling; clinicians appropriate use and error reporting; and model providers disclosure of changes and limitations [6]. Several want clinicians, and sometimes institutions, spared default or sole liability, especially for outputs they cannot check [7]; one warns providers will otherwise not adopt the tools [8].

Institutions' capacity divides commenters. One argues most health systems cannot inventory their software devices or record model versions, so their role should be receiving and reporting, and warns that relying on them would leave patients at safety-net and rural sites less protected [9]; another wants minimum site capacity for higher-tier functions, coordinated with CMS conditions of participation [10]. Three would not ask institutions to validate or independently evaluate the model [11], while others see local validation as a needed or appropriate safety layer, since manufacturer testing does not establish performance everywhere [12]. As larger changes, one wants institutions to show regulatory expertise, quality systems and a monitoring protocol before deploying [13]; another wants manufacturers required to let institutions list, limit, log and stop a device [14]. Two ask FDA to clarify when an institution's

configuration or monitoring creates a modification or makes it a manufacturer [15].

Roles proposed for others include professional societies writing reviewer calibration materials, failure taxonomies, performance measures and reviewer qualifications [16]; standards bodies setting common log, attribution and incident formats [17]; health plans supplying samples and outcome data under contract [18]; and clinical engineering and service staff, whom one says the paper omits [19]. One commenter wants society participation kept voluntary and funded [20].

Several want easy routes for clinicians, patients and caregivers to report suspected problems [21], one with timely replies on what was found [22]. One wants institutions told of model changes in a set format and timing [23]; two would pool signals across manufacturers or sites [24]; another wants independent data systems governed with professional societies, funded without buying favorable findings [25]. Process ideas include a role matrix for who may pause, investigate and approve return to service [26], cross-organization handoffs rehearsed within the hazard's response time [27], and a drafting test asking whether each duty names its bearer [28].

**Sources.** [1] 0003 ¶p057; 0005 ¶p325; 0009 ¶p043; 0013 ¶p126; 0017 ¶p039; 0032 ¶p101; 0033 ¶p035; 0035 ¶p078; 0042 ¶p054; 0045 ¶p084; 0049 ¶p085; 0050 ¶p017; 0052 ¶p133; 0055 ¶p240; 0056 ¶p025; 0075 ¶p026; 0076 ¶p037; 0083 ¶p005; 0086 ¶p009; 0087 ¶p041; 0092 ¶p009; 0093 ¶p030; 0099 ¶p015; 0100 ¶p063; 0105 ¶p032; 0113 ¶p039; 0116 ¶p039; 0118 ¶p060; 0119 ¶p037; 0120 ¶p245; 0127 ¶p040; 0130 ¶p012 [2] 0013 ¶p126 [3] 0099 ¶p016 [4] 0025 ¶p258, ¶p259 [5] 0003 ¶p057; 0017 ¶p007; 0025 ¶p247, ¶p263; 0045 ¶p084; 0047 ¶p012; 0049 ¶p084; 0075 ¶p027; 0081 ¶p001; 0087 ¶p041; 0092 ¶p009; 0097 ¶p013; 0105 ¶p032; 0113 ¶p039; 0120 ¶p245; 0121 ¶p053; 0130 ¶p012 [6] 0017 ¶p051; 0055 ¶p241, ¶p242, ¶p243; 0120 ¶p119, ¶p120, ¶p121, ¶p122; 0121 ¶p053 [7] 0017 ¶p051; 0076 ¶p037; 0083 ¶p005; 0097 ¶p008; 0113 ¶p039; 0120 ¶p245; 0121 ¶p053 [8] 0113 ¶p003 [9] 0050 ¶p002, ¶p007, ¶p013, ¶p017 [10] 0094 ¶p030 [11] 0050 ¶p017; 0119 ¶p039; 0130 ¶p012 [12] 0052 ¶p090; 0087 ¶p020; 0120 ¶p107 [13] 0113 ¶p020 [14] 0130 ¶p001, ¶p015 [15] 0035 ¶p035; 0100 ¶p063 [16] 0003 ¶p057; 0033 ¶p033; 0055 ¶p244; 0056 ¶p025; 0119 ¶p040; 0120 ¶p243 [17] 0042 ¶p054; 0055 ¶p245; 0093 ¶p030; 0120 ¶p244 [18] 0009 ¶p043 [19] 0032 ¶p103, ¶p104 [20] 0118 ¶p061 [21] 0025 ¶p233; 0033 ¶p034; 0056 ¶p026; 0075 ¶p028; 0118 ¶p055 [22] 0118 ¶p055 [23] 0050 ¶p011 [24] 0021 ¶p018; 0055 ¶p246 [25] 0116 ¶p029, ¶p041 [26] 0105 ¶p032 [27] 0075 ¶p027, ¶p030 [28] 0119 ¶p006

## Q22: Scaling evidence for modifications; what a PCCP can cover

40 comments, 80 tags: 2 support, 76 conditional-minor, 1 conditional-major, 1 neutral

Many accept the premarket assessment as the baseline for changes, with evidence scaled to each change; 76 of the 80 points ask for small changes [1]. Many propose tiers: changes that cannot affect clinical output, such as interface or formatting, handled in the quality system; bounded prompt, retrieval or guardrail changes re-tested, often under a PCCP; and new uses, populations or autonomy sent to FDA [2]. Model updates divide them: one would keep updates within a model family in the quality system after full re-benchmarking, sending a new family or provider to FDA [3]; another sends any core architecture change to a new submission [4]; a third would exclude open-ended foundation model changes from PCCPs [5].

Several judge a change by its effect on behavior, not its technical size, since a small edit can alter triage or escalation [6]. Several would map each change type in advance to the benchmark elements it could affect [7], and several would always re-run a core of safety-critical tests after changes that could affect output or safety behavior [8]. Others want change categories fitting GenAI components, with prompt, retrieval and orchestration changes recognized as device changes [9], and model compression named [10]; one found no public summary of an authorized PCCP, mostly for non-generative devices, covering such components [11].

To keep comparisons sound, commenters want benchmark datasets versioned [12], the locked suite never revised alongside the device without FDA review [13], automated adjudicators versioned and bridge-tested when replaced [14], and results compared by failure mode, not element totals [15]. One

asks sponsors to show their evaluation's own run-to-run noise, for instance by resampling the unchanged configuration against itself, so thresholds neither mistake noise for change nor block most changes [16].

Several tie evidence to the whole deployed configuration, not the model alone, so a change to any part prompts assessment or re-testing [17]; one wants deployers told before changes take effect [18]. PCCPs, some say, should state what may change, its limits, tests, acceptance criteria and rollback [19], with changes outside the plan returned for review [20]. As a larger change, one would bind evidence to an immutable configuration, any change triggering reassessment [21].

**Sources.** [1] 0012 ¶p011; 0013 ¶p130; 0039 ¶p005; 0045 ¶p085; 0049 ¶p087; 0052 ¶p135; 0055 ¶p248; 0058 ¶p043; 0071 ¶p031; 0074 ¶p006; 0087 ¶p031; 0093 ¶p031; 0105 ¶p034; 0113 ¶p040; 0116 ¶p082; 0118 ¶p066; 0120 ¶p247 [2] 0003 ¶p060, ¶p061, ¶p062; 0005 ¶p327; 0013 ¶p131, ¶p132, ¶p133; 0045 ¶p086; 0049 ¶p087; 0055 ¶p249, ¶p250, ¶p252, ¶p253; 0087 ¶p032, ¶p033; 0093 ¶p031; 0105 ¶p035; 0113 ¶p040; 0118 ¶p066; 0120 ¶p248, ¶p250, ¶p251 [3] 0013 ¶p132, ¶p133 [4] 0003 ¶p060 [5] 0118 ¶p066 [6] 0013 ¶p130; 0025 ¶p353; 0035 ¶p082; 0045 ¶p085; 0052 ¶p135; 0067 ¶p032; 0071 ¶p031 [7] 0035 ¶p082; 0067 ¶p032; 0090 ¶p035; 0091 ¶p056; 0114 ¶p033 [8] 0005 ¶p299; 0058 ¶p043; 0071 ¶p032; 0091 ¶p057; 0118 ¶p066 [9] 0090 ¶p035; 0099 ¶p021 [10] 0027 ¶p036 [11] 0090 ¶p033 [12] 0085 ¶p035; 0090 ¶p036 [13] 0013 ¶p135 [14] 0091 ¶p059 [15] 0027 ¶p051 [16] 0091 ¶p021, ¶p033 [17] 0075 ¶p032; 0088 ¶p024; 0114 ¶p032; 0127 ¶p036 [18] 0130 ¶p013 [19] 0024 ¶p010; 0055 ¶p254, ¶p256, ¶p257, ¶p258, ¶p261; 0120 ¶p251 [20] 0055 ¶p262; 0105 ¶p035 [21] 0076 ¶p047

## Q23: Adapting PCCPs to changes that can't be fully prespecified

19 comments, 29 tags: 18 conditional-minor, 10 conditional-major, 1 neutral

Most would shift what a PCCP prespecifies from the changes themselves to what must hold after any change, such as the evaluation suite, acceptance thresholds, allowed deviation from baseline, verification steps and rollback; of the 29 points, 18 ask for small changes and 10 for larger ones [1]. Changes that pass could then proceed without a new submission, while failures are rolled back or submitted [2]. One stresses that this works only if the benchmark suite is escrowed and any change to it treated as a new submission [3]; another would also fix safety properties, such as escalation and human override, that cannot lapse without revalidation [4].

Three warn that the plan must stay bounded, not an open-ended authorization [5]: stable averages can hide decline in a subgroup or safety-critical task [6], and changes to intended use or autonomy need ordinary review [7].

Two propose tiers [8]: one by size of accuracy change, overall and by subgroup, from no submission to expedited or full review [9]; the other keeping narrow, fully specified changes under traditional PCCPs and governing broader change by a pre-authorized re-benchmarking protocol [10].

On learning in use, one wants corrections used offline to build validated new versions, never silent learning in production [11]; another wants a defined, evaluated scope for memory or adaptive retrieval [12]. One would let PCCPs move devices from supervised to more independent use as criteria are met, and back on postmarket signals [13].

**Sources.** [1] 0005 ¶p328; 0013 ¶p138; 0016 ¶p004; 0025 ¶p358; 0035 ¶p083; 0045 ¶p088; 0049 ¶p089; 0051 ¶p024; 0052 ¶p137; 0058 ¶p042; 0087 ¶p034; 0093 ¶p032; 0105 ¶p035; 0113 ¶p041; 0118 ¶p068; 0120 ¶p253 [2] 0005 ¶p328; 0013 ¶p145; 0016 ¶p004 [3] 0013 ¶p147, ¶p148 [4] 0025 ¶p359, ¶p361 [5] 0105 ¶p035; 0116 ¶p045; 0120 ¶p255 [6] 0116 ¶p045 [7] 0120 ¶p256 [8] 0113 ¶p041; 0118 ¶p068 [9] 0113 ¶p041 [10] 0118 ¶p068 [11] 0120 ¶p112 [12] 0075 ¶p035 [13] 0052 ¶p138

## Q24: Changes initiated by third-party foundation model developers

45 comments, 104 tags: 1 support, 100 conditional-minor, 3 conditional-major

Several hold the manufacturer answerable for upstream changes it did not make, since a changed

model cannot excuse the device [1]. Most want contractual and technical controls combined; 100 of the 104 points ask for small changes [2].

Requested contract terms include advance notice, version and release information, incident disclosure, deprecation windows, access to earlier versions and rollback support [3]; proposed notice runs from 14 days with sandbox access [4] to 90 days [5], or long enough to re-benchmark [6]. Several find notice insufficient on its own: it states the vendor's intent, not the effect on the device, cannot be verified in real time, may not match the component actually running, and small manufacturers cannot compel it [7].

Many want version pinning [8]; as larger changes, two would make it a condition of clearance and keep PCCPs from covering unpinned or uncontrolled changes [9]. One notes pinning is not always offered, ends in forced migration and misses changes under a stable label [10]; another wants forced migration treated as a major change, with a validated fallback ready [11]. Two would sidestep the problem with locked on-device or private models [12].

Many want detection that does not depend on the provider, such as fixed canary probes run against the live model, output drift monitoring and configuration fingerprints [13]. One warns that manufacturers may not know which model is running, and wants each output to carry a machine-generated record of what served it [14]; another would count deployer-observed changes as evidence alongside vendor notices [15].

Many want no new version in clinical use until it is re-tested, often through shadow runs, quarantine or staged release, and no silent replacement [16]. Several add that a provider refusing pinning or notice leaves unresolved risk, to be met by narrower use, more evidence, disclosure or another supplier [17]. Four see PCCPs as suited to anticipated upgrades that pass prespecified re-testing [18].

One would keep safety-critical logic outside the third-party model, so upstream change mainly affects communication [19]; another asks FDA not to require self-hosting, model weights, indefinite old-version support or one notice period for all devices [20]; a third wants plans for a provider retiring a model or failing outright [21].

**Sources.** [1] 0003 ¶p064; 0017 ¶p041; 0021 ¶p019; 0025 ¶p265; 0032 ¶p057; 0045 ¶p089; 0055 ¶p279; 0099 ¶p017; 0100 ¶p012; 0113 ¶p042; 0126 ¶p214 [2] 0003 ¶p065; 0005 ¶p329; 0006 ¶p027; 0013 ¶p153; 0017 ¶p041; 0018 ¶p038; 0025 ¶p267; 0030 ¶p013; 0032 ¶p057; 0035 ¶p083; 0045 ¶p090; 0048 ¶p009; 0049 ¶p090; 0055 ¶p264, ¶p272; 0090 ¶p038; 0093 ¶p033; 0100 ¶p012; 0105 ¶p036; 0113 ¶p042; 0116 ¶p046; 0118 ¶p070; 0120 ¶p259; 0123 ¶p062; 0126 ¶p213; 0130 ¶p013 [3] 0017 ¶p041; 0035 ¶p083; 0055 ¶p264, ¶p267, ¶p268, ¶p269, ¶p271; 0113 ¶p042; 0120 ¶p259 [4] 0102 ¶p010 [5] 0003 ¶p064 [6] 0013 ¶p152; 0123 ¶p062 [7] 0006 ¶p027; 0013 ¶p153; 0062 ¶p050; 0090 ¶p039; 0131 ¶p061 [8] 0003 ¶p065; 0006 ¶p027; 0013 ¶p151; 0018 ¶p037; 0032 ¶p057; 0035 ¶p083; 0048 ¶p009; 0049 ¶p091; 0053 ¶p010; 0055 ¶p273; 0062 ¶p042; 0076 ¶p044; 0090 ¶p040; 0100 ¶p066; 0105 ¶p036; 0116 ¶p046; 0120 ¶p259; 0123 ¶p062; 0126 ¶p213; 0130 ¶p013 [9] 0062 ¶p041, ¶p042; 0076 ¶p044 [10] 0090 ¶p040 [11] 0013 ¶p154 [12] 0082 ¶p034; 0096 ¶p007 [13] 0005 ¶p329; 0013 ¶p153; 0030 ¶p014; 0035 ¶p083; 0048 ¶p009; 0054 ¶p104; 0055 ¶p276; 0090 ¶p041; 0093 ¶p033; 0096 ¶p011; 0118 ¶p070; 0124 ¶p012; 0127 ¶p036; 0131 ¶p061 [14] 0062 ¶p015, ¶p040 [15] 0131 ¶p058 [16] 0003 ¶p065; 0006 ¶p027; 0049 ¶p091; 0054 ¶p105; 0055 ¶p275; 0076 ¶p044; 0093 ¶p033; 0100 ¶p067; 0102 ¶p010; 0105 ¶p036; 0118 ¶p070; 0120 ¶p262; 0126 ¶p213 [17] 0003 ¶p065; 0013 ¶p151; 0021 ¶p019; 0048 ¶p009; 0055 ¶p280; 0075 ¶p038; 0120 ¶p261 [18] 0006 ¶p027; 0100 ¶p068; 0120 ¶p260; 0126 ¶p214 [19] 0100 ¶p065 [20] 0075 ¶p041 [21] 0032 ¶p066, ¶p067

## MASTER\_FILES: Foundation model master files

39 comments, 107 tags: 14 support, 70 conditional-minor, 18 conditional-major, 5 neutral

### Q25: Voluntary Foundation Model Master Files

39 comments, 107 tags: 14 support, 70 conditional-minor, 18 conditional-major, 5 neutral

No point opposes the program, 70 of 107 ask for small changes and 18 for larger ones; many stress a master file cannot replace the device maker's responsibility for, or evidence on, its finished device [1]. Several add, as the paper does, that filing is not FDA approval of the model or a device [2].

Several expect files to cut duplicated review or diligence [3], and several tie their value to staying current and version-specific [4]. Many list contents, such as version history, known limitations and failure modes, training-data provenance, subgroup results, guardrails and change-notice commitments [5]. Some add deprecation policies [6], rollback options [7] or provider-exit plans [8]; one adds refusal behavior and policy changes between versions, which sponsors cannot assess from outside [9].

Many doubt developers have reason to disclose weaknesses or keep files current [10]. Two would make disclosure mandatory [11], one for higher-consequence devices [12], the other refusing authorization when a developer will not disclose [13]; another would require it unless the sponsor carries the full evidence burden itself [14]. Two would put the cost of a missing file on the device [15]: more premarket evidence [16], or the manufacturer characterizing the model fully, so that manufacturers press suppliers to file [17]. One asks FDA to consider expecting a file or equivalent when a sponsor relies materially on a third-party model [18]; another would have sponsors document the model's identity, version and change behavior even without a file [19]. By contrast, one wants no presumption against devices lacking a file [20], another a strictly voluntary, non-gating program [21]. Three suggest rewards for filing [22]: more predictable review [23], or meeting part of benchmarking or post-update re-examination by reference [24].

Three see limits that willingness cannot fix [25]. One notes developers have missed their own models' deviations, and asks CDRH to consider requiring filers to disclose their detection methods [26]. Another notes a file describes the model as released, not as fine-tuned and configured in the device, and wants submissions to document such changes [27]. A third measured, on one workstation over one day, a hosted assistant's answering model being swapped 153 times in fourteen hours, so "currency, not candour" binds [28]; it would limit files to stable matters such as model family and notice commitments [29], add execution locality and a version identifier the serving system reports [30], and record which model answered, on the user's side or inside the device [31].

One traces reluctance mostly to records leaving the developer's control, and would build files on hash commitments to specific results [32], expecting most from models others cannot re-test [33]. Three value files that keep proprietary details protected [34]; as larger changes, one wants a narrow public tier, such as authorized versions, change dates and safety events [35], and another wants model cards for marketed devices public, since characterized limitations "are labeling" [36]. Alternatives proposed include contractual disclosure, audit rights, independent attestations or audits, standardized model cards and machine-readable version information [37]; supplier requirements under purchasing controls [38], and cloud platforms as file holders [39].

**Sources.** [1] 0013 ¶p164; 0025 ¶p366, ¶p368; 0027 ¶p068; 0032 ¶p102; 0046 ¶p020; 0049 ¶p095; 0054 ¶p096; 0055 ¶p119, ¶p132; 0087 ¶p038; 0099 ¶p017; 0105 ¶p038; 0114 ¶p026; 0115 ¶p043; 0119 ¶p055; 0120 ¶p268; 0126 ¶p217 [2] 0046 ¶p020; 0054 ¶p096; 0087 ¶p038; 0120 ¶p122; 0126 ¶p216 [3] 0025 ¶p364; 0049 ¶p093; 0055 ¶p133; 0087 ¶p036; 0100 ¶p070; 0105 ¶p037; 0115 ¶p042; 0119 ¶p055 [4] 0005 ¶p330; 0013 ¶p163; 0017 ¶p043; 0035 ¶p084; 0049 ¶p093; 0105 ¶p037; 0120 ¶p267 [5] 0003 ¶p070; 0005



¶p330; 0006 ¶p027; 0013 ¶p162; 0017 ¶p043; 0035 ¶p085; 0049 ¶p094; 0054 ¶p096; 0055 ¶p120, ¶p123, ¶p128; 0093 ¶p034; 0105  
¶p037; 0115 ¶p042; 0120 ¶p265, ¶p266 [6] 0013 ¶p162; 0032 ¶p102; 0035 ¶p085; 0054 ¶p096; 0120 ¶p265 [7] 0035 ¶p085; 0055  
¶p130; 0115 ¶p042 [8] 0032 ¶p102; 0055 ¶p131 [9] 0058 ¶p045 [10] 0003 ¶p068; 0013 ¶p158; 0017 ¶p043; 0045 ¶p092; 0076  
¶p039; 0093 ¶p034; 0100 ¶p071; 0102 ¶p010; 0106 ¶p018; 0113 ¶p044; 0115 ¶p043; 0118 ¶p072; 0119 ¶p055 [11] 0106 ¶p018; 0113  
¶p044 [12] 0106 ¶p018 [13] 0113 ¶p044 [14] 0076 ¶p040 [15] 0003 ¶p069; 0013 ¶p160 [16] 0003 ¶p069 [17] 0013 ¶p160,  
¶p161 [18] 0017 ¶p043 [19] 0093 ¶p034 [20] 0100 ¶p071 [21] 0094 ¶p025 [22] 0006 ¶p027; 0058 ¶p047; 0118 ¶p072 [23] 0006  
¶p027 [24] 0058 ¶p047; 0118 ¶p072 [25] 0008 ¶p010; 0027 ¶p067; 0079 ¶p042 [26] 0008 ¶p010, ¶p011 [27] 0027 ¶p067, ¶p070  
[28] 0079 ¶p018, ¶p022 [29] 0079 ¶p032 [30] 0079 ¶p035, ¶p038 [31] 0079 ¶p046, ¶p064 [32] 0131 ¶p064, ¶p065 [33] 0131  
¶p068 [34] 0012 ¶p010; 0035 ¶p085; 0054 ¶p008 [35] 0050 ¶p016 [36] 0076 ¶p041 [37] 0049 ¶p096; 0093 ¶p034; 0115 ¶p043;  
0120 ¶p269 [38] 0119 ¶p052 [39] 0100 ¶p072

## AGENTIC: Agentic AI systems

53 comments, 182 tags: 5 support, 146 conditional-minor, 21 conditional-major, 10 neutral

### Q26: Agentic AI systems

53 comments, 182 tags: 5 support, 146 conditional-minor, 21 conditional-major, 10 neutral

No point opposes FDA's approach and 146 of 182 ask for small changes; many stress that because agents act, evaluation must cover the whole action sequence, not only the final output [1]. Several note that errors compound over steps [2]; another notes that individually correct steps can add up to delayed or abandoned care [3].

About half want human approval, usually before irreversible or high-consequence actions [4], several unless autonomy has been specifically evaluated or justified [5]. Several add that a checkpoint must be meaningful: not bypassable, open to argument or satisfiable by another model, and giving the reviewer enough information [6]. As larger changes, one would redefine oversight of irreversible actions as authorization before the act, not review after [7]; another wants proof, a tamper-evident record that a present, uncoerced person authorized each consequential action, shown premarket [8]; a third wants checkpoints required for physical outputs such as printer build files [9].

Many want each agent's authority declared and enforced, such as permitted actions, tools and data, least privilege, and denial of anything else [10]; several want limits enforced outside the model rather than by prompts or instructions [11]. Several separate the ability to plan or act from the authority to do so [12].

Many want ways to stop or contain an agent and return to a safe state, such as a kill switch, rollback or handoff to a human when tools fail [13]. Many want audit records of agents' actions, often full enough to reconstruct what happened [14], several protected against tampering [15]. Three warn that an agent's account of its own steps is unverified [16], one adding it may be the only trace [17]; two want records to show who acted and who authorized, and to flag skipped checkpoints and unattributed actions [18].

Several would scale controls, evidence or acceptance criteria to an agent's autonomy, authority or the reversibility of its actions [19]. Proposed criteria include task-level rather than per-step success [20], per-action floors with the highest-consequence actions measured most carefully [21], safe recovery as a pass-fail competency [22], rates of prohibited actions and unauthorized tool calls [23], and, as a larger change, withholding a capability until an authorization bypass is fixed [24]. Element A.1 draws mixed views: one calls it "woefully inadequate" [25]; two support its competencies [26], one listing gaps such as whether an agent's step reports are true [27]; another would test whether an agent notices its authority expanding mid-task [28].

Three want non-device functions such as scheduling or documentation to stay outside device rules when agents orchestrate them [29], while one warns administrative agents can shape care outside device rules, suggesting FDA, CMS, states and others may need to coordinate [30]. One would judge agents controlling other devices against the controlled device's risk [31]. One would presume agentic devices ineligible for reduced premarket evidence [32]; another wants prospective clinical validation for agents that treat, prescribe or judge the need for emergency care [33]; a third, more frequent postmarket oversight and reassessment after any material change in the action chain [34]. As a larger change, one would register and version each agent, model, tool and qualifying dataset, blocking any change until

prespecified evidence exists [35].

**Sources.** [1] 0005 ¶p331; 0018 ¶p030; 0025 ¶p275; 0035 ¶p087; 0048 ¶p010; 0049 ¶p097; 0052 ¶p145; 0056 ¶p027; 0071 ¶p028; 0075 ¶p011; 0087 ¶p047; 0105 ¶p039; 0113 ¶p045; 0115 ¶p044 [2] 0013 ¶p166; 0035 ¶p089; 0049 ¶p099; 0052 ¶p145; 0055 ¶p136; 0056 ¶p027; 0067 ¶p033; 0095 ¶p046; 0118 ¶p075 [3] 0025 ¶p276, ¶p282 [4] 0003 ¶p073; 0013 ¶p168; 0017 ¶p045; 0018 ¶p031; 0035 ¶p089; 0046 ¶p020; 0048 ¶p010; 0049 ¶p097; 0052 ¶p148; 0054 ¶p099; 0055 ¶p147; 0056 ¶p028; 0067 ¶p033; 0076 ¶p063; 0087 ¶p046; 0089 ¶p008; 0092 ¶p019; 0093 ¶p035; 0097 ¶p006; 0098 ¶p045; 0100 ¶p074; 0102 ¶p010; 0105 ¶p040; 0113 ¶p046; 0115 ¶p046; 0118 ¶p075; 0120 ¶p273 [5] 0035 ¶p089; 0054 ¶p099; 0087 ¶p046; 0105 ¶p040; 0120 ¶p273 [6] 0048 ¶p010; 0056 ¶p028; 0067 ¶p033; 0075 ¶p013; 0094 ¶p068; 0098 ¶p045; 0100 ¶p074; 0102 ¶p010; 0111 ¶p021 [7] 0003 ¶p073 [8] 0089 ¶p002, ¶p006 [9] 0053 ¶p012 [10] 0013 ¶p169; 0017 ¶p045; 0024 ¶p008; 0035 ¶p088; 0048 ¶p010; 0055 ¶p144, ¶p152; 0056 ¶p028; 0075 ¶p012; 0076 ¶p063; 0093 ¶p035; 0101 ¶p006; 0105 ¶p040; 0115 ¶p046; 0130 ¶p014 [11] 0035 ¶p089; 0061 ¶p055; 0075 ¶p012; 0093 ¶p035; 0127 ¶p042 [12] 0029 ¶p018; 0049 ¶p103; 0087 ¶p045; 0092 ¶p019; 0122 ¶p025 [13] 0003 ¶p075; 0017 ¶p045; 0018 ¶p032; 0022 ¶p007; 0024 ¶p008; 0035 ¶p089; 0048 ¶p010; 0052 ¶p148; 0055 ¶p150, ¶p151; 0056 ¶p029; 0100 ¶p076; 0105 ¶p040; 0109 ¶p005; 0113 ¶p046; 0114 ¶p009; 0115 ¶p046; 0118 ¶p076; 0120 ¶p272; 0127 ¶p042; 0130 ¶p014 [14] 0013 ¶p170; 0017 ¶p045; 0018 ¶p031; 0049 ¶p099; 0054 ¶p099; 0055 ¶p148; 0056 ¶p029; 0067 ¶p034; 0075 ¶p048; 0076 ¶p063; 0093 ¶p035; 0113 ¶p046; 0115 ¶p045; 0120 ¶p272; 0130 ¶p014 [15] 0017 ¶p045; 0067 ¶p033; 0075 ¶p049; 0076 ¶p063; 0089 ¶p006; 0122 ¶p012; 0127 ¶p042 [16] 0061 ¶p009; 0075 ¶p051; 0127 ¶p042 [17] 0061 ¶p014 [18] 0042 ¶p047, ¶p048; 0131 ¶p072, ¶p073 [19] 0027 ¶p079; 0045 ¶p094; 0049 ¶p097; 0051 ¶p025; 0052 ¶p148; 0054 ¶p009; 0087 ¶p050; 0089 ¶p017; 0093 ¶p035; 0097 ¶p006; 0118 ¶p075 [20] 0013 ¶p166 [21] 0095 ¶p046 [22] 0114 ¶p009 [23] 0115 ¶p047 [24] 0075 ¶p017 [25] 0043 ¶p018 [26] 0061 ¶p042; 0093 ¶p035 [27] 0061 ¶p044 [28] 0071 ¶p034 [29] 0094 ¶p069, ¶p070; 0100 ¶p073; 0120 ¶p275 [30] 0025 ¶p287, ¶p294, ¶p296 [31] 0051 ¶p025 [32] 0013 ¶p172 [33] 0097 ¶p006 [34] 0056 ¶p030 [35] 0122 ¶p007, ¶p008, ¶p014